Intraoral Anatomy
Dr.Manila

General principles of radiographic evaluation

- The radiographic appearance of structures reflects the pattern of x-ray photon attenuation
- High-attenuate tissues appear more radiopaque (brighter)
  - Enamel, cortical bone, amalgam restoration
- Low-attenuation tissues appear radiolucent (darker)
  - Pulp canal, maxillary sinus, airway space
- 2-dimensional projections structures along the path of beam superimpose
- CT –visualization of anatomy in 3-dimensions
- Diagnostic interest may be limited to a specific site, but important to evaluate all structures imaged

Types of Bone- Cortical

- *Cortex* = "outer layer"
- Aka Compact Bone
- Dense
- Radiopaque
- Example: inferior border of the mandible

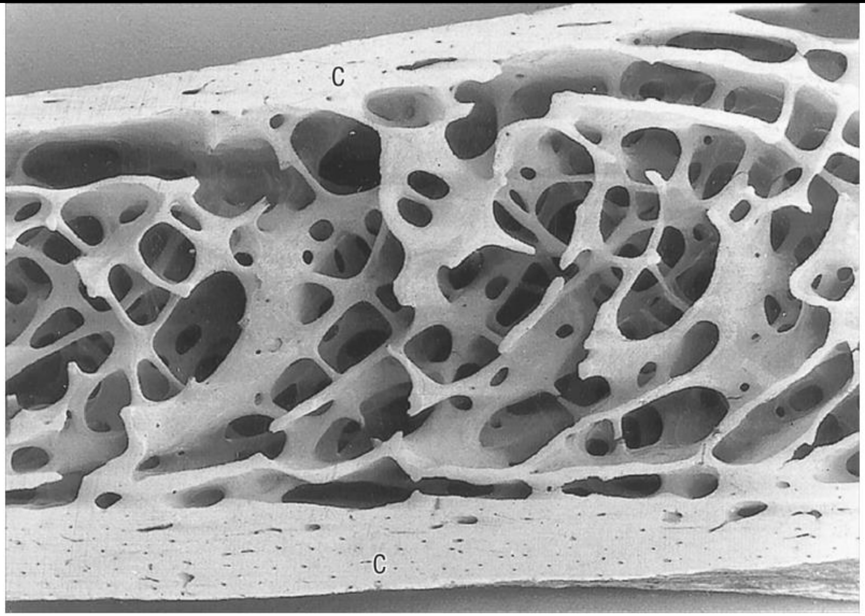


FIG 28-1 Cortical bone and cancellous bone. (From Standrings: Gray's anatomy, ed 40, London, 2009, Churchill Livingstone.)

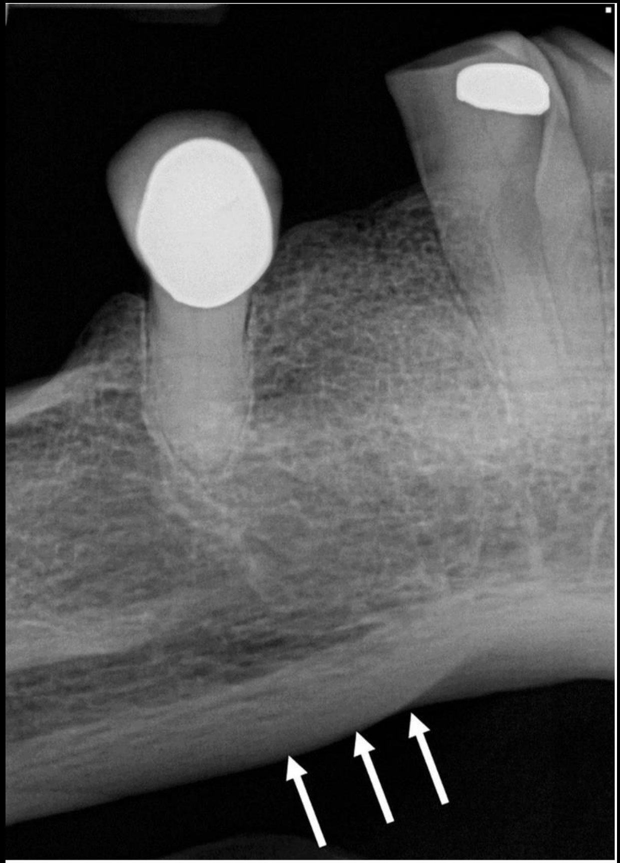


FIG 28-2 Cortical bone appears highly radiopaque on a dental image.

Types of Bone- Cancellous

- *Cancelli* = "crossbars"; "arranged like a lattice"
- Soft, Spongy Bone
- Trabeculae are radiopaque, otherwise radiolucent
- Example: alveolar bone

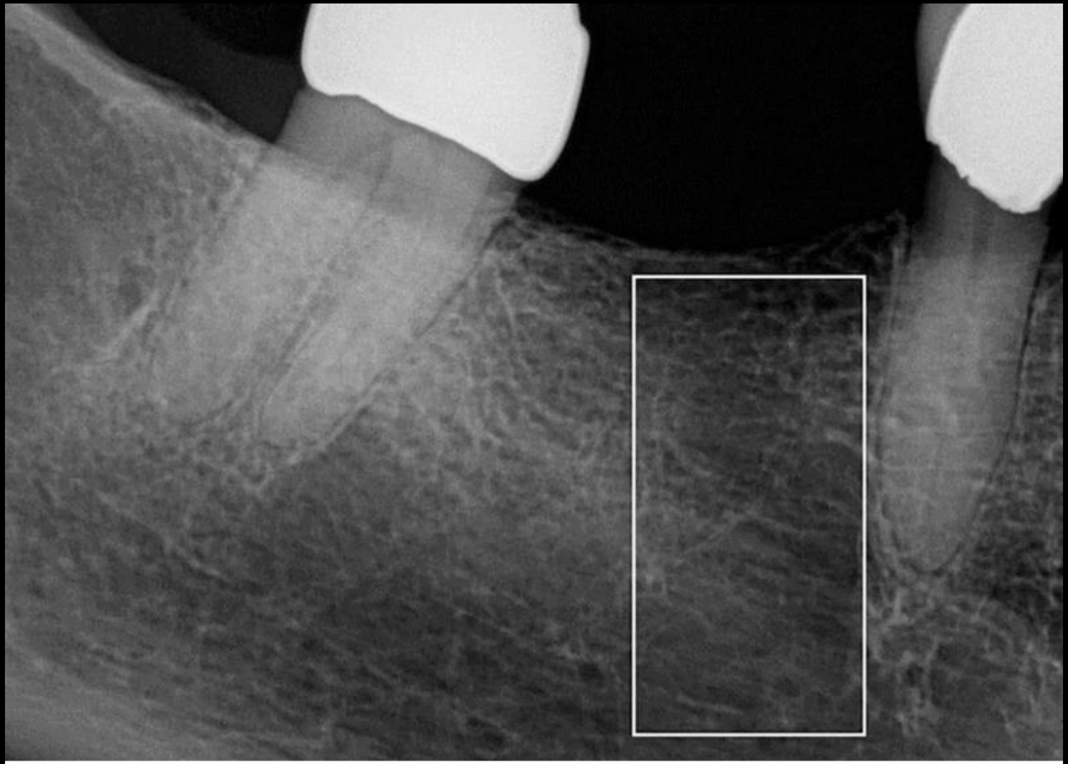
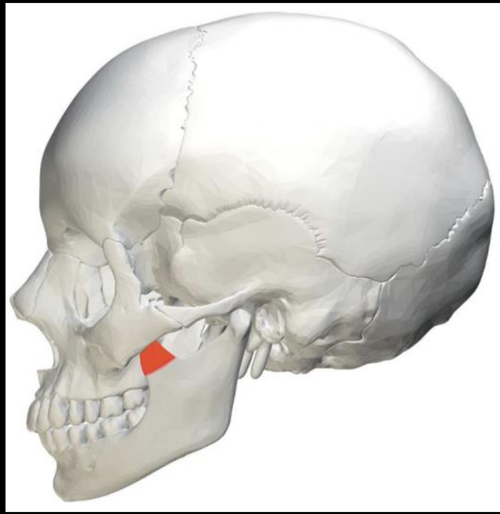
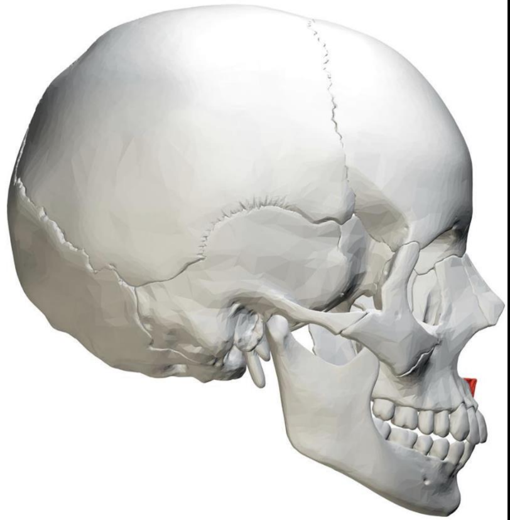


FIG 28-3 Cancellous bone appears predominantly radiolucent.

Bony Prominences

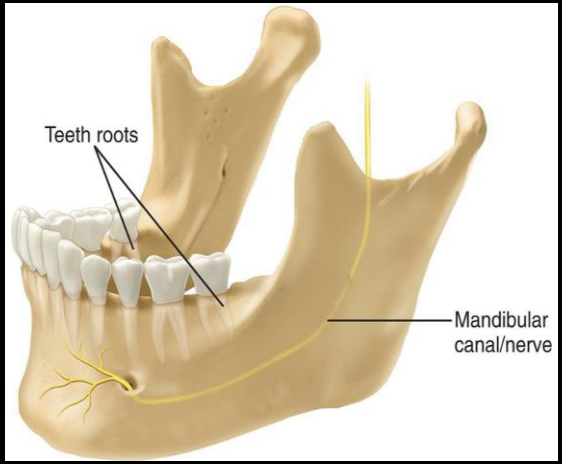
- Dense cortical bone
- Radiopaque
- Described as:
  - Process, ridge, spine, tubercle, tuberosity

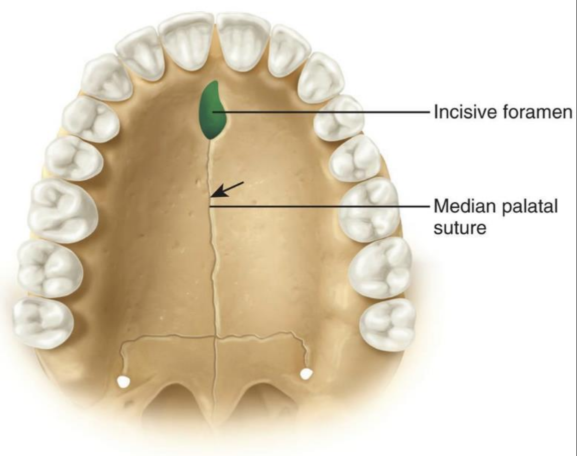


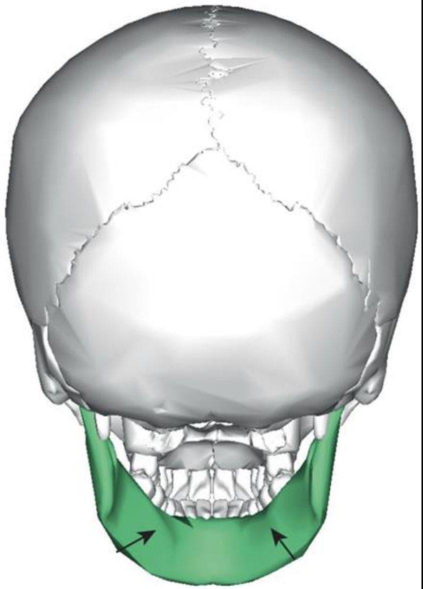


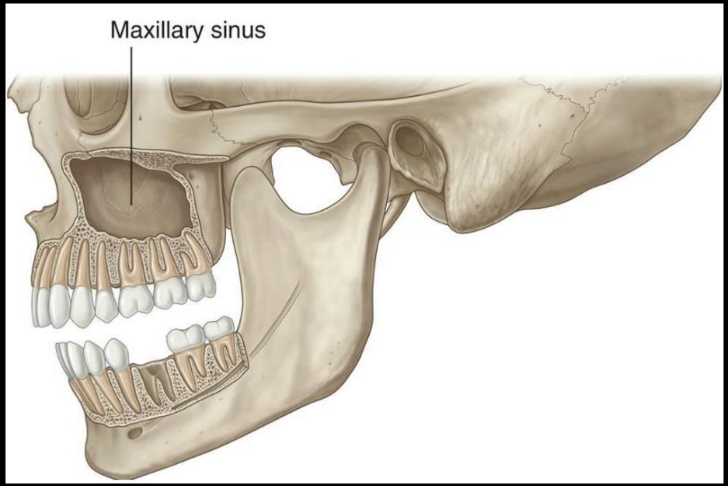

Spaces and Depressions

- Radiolucent
- Described as:
  - Canal, foramen, fossa, sinus





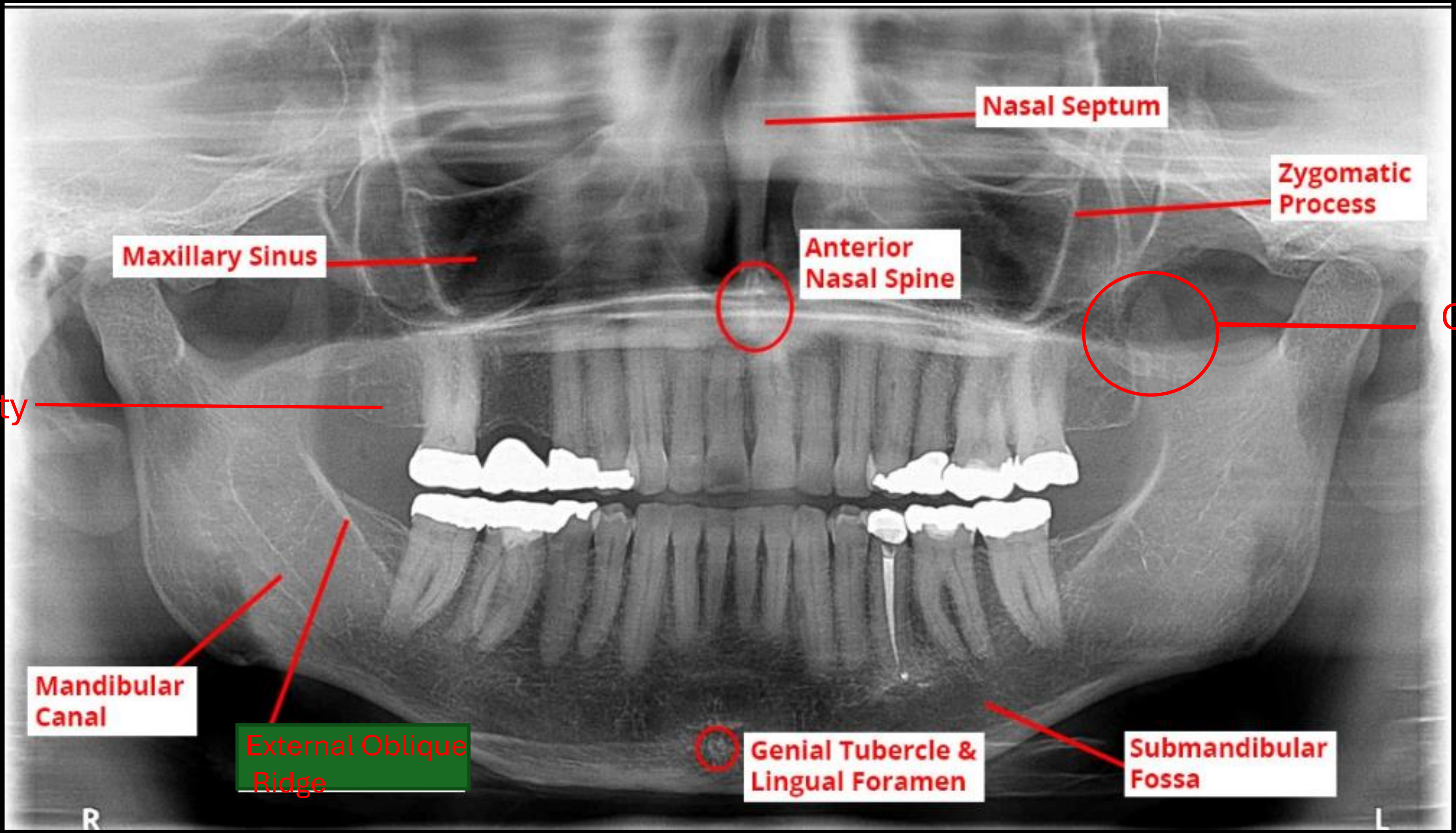




General Terms

- Septum- bony wall
- **Suture- junction where two skull bones** meet

Radiographic Images



Normal Anatomic Landmarks-Maxilla

- Incisive Foramen
- Superior foramina of the incisive canal
- Median palatal suture
- Lateral fossa
- Nasal cavity
- Nasal septum
- Floor of nasal cavity

- Anterior nasal spine
- Inferior nasal conchae
- Maxillary sinus
- Nutrient canals
- Inverted Y
- Maxillary tuberosity
- Hamulus
- Zygomatic process
- Zygoma

Incisive Foramen

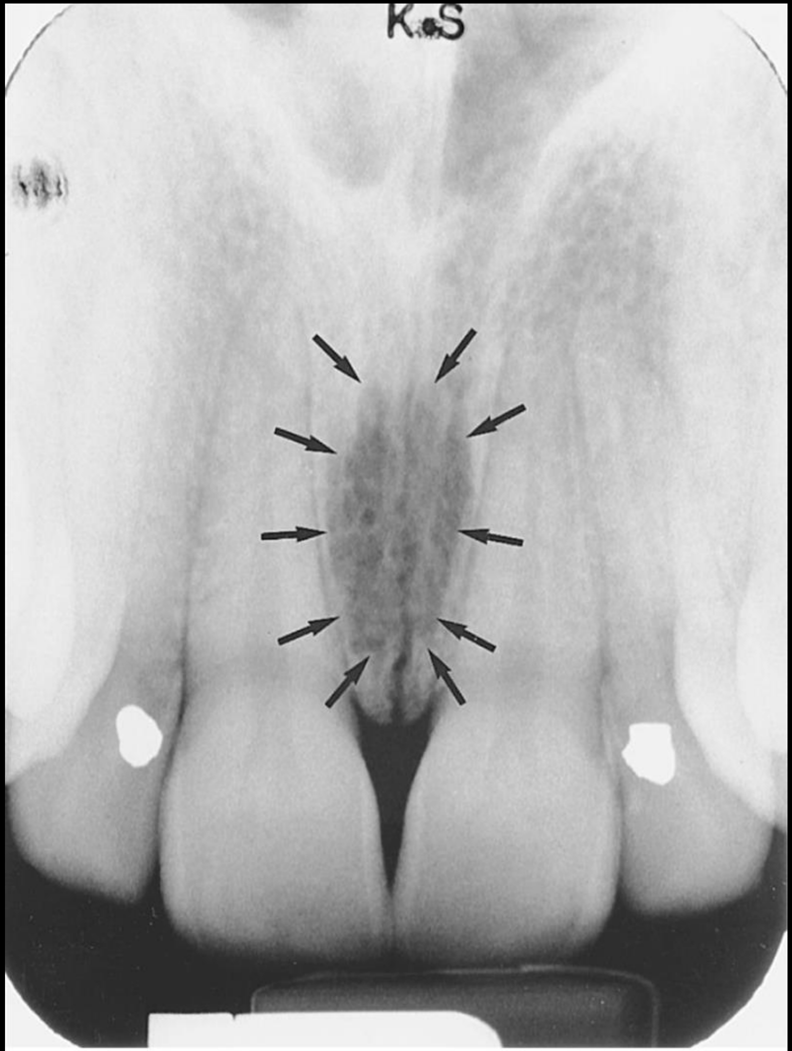


FIG 28-14 The incisive foramen appears radiolucent. (From Haring JI, Lind LJ: Radiographic interpretation for the dental hygienist, Philadelphia, 1993, Saunders.)

Superior foramina of the incisive canal

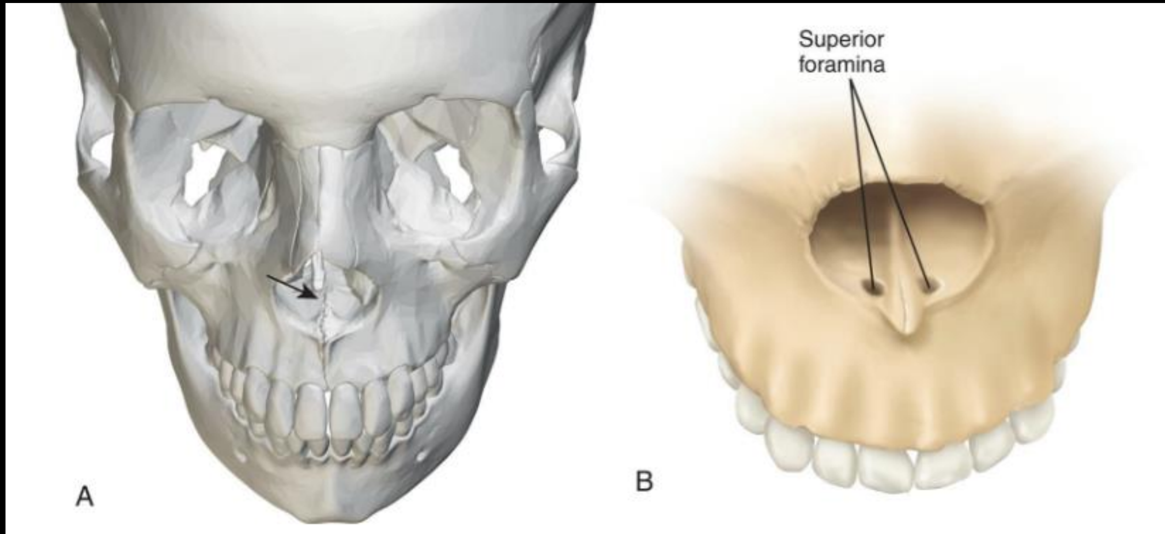
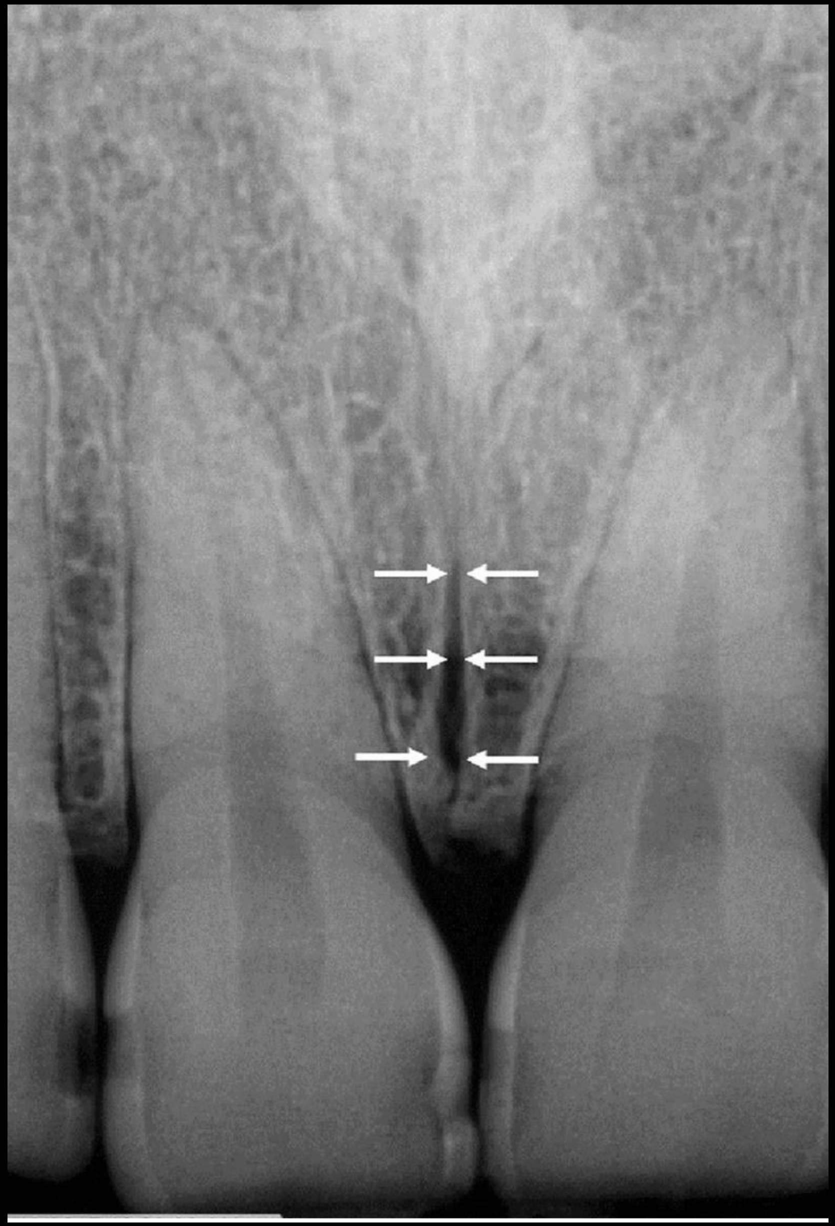


FIG 28-15 A, The superior foramina of the incisive canal are located on the anterior floor of the nasal cavity. (Images created from BodyParts3D, © The Database Center for Life Science licensed under CC Attribution-Share Alike 2.1 Japan [http://creativecommons.org/licenses/by-sa/2.1/jp/legalcode] / Modified from User: Was a bee / Wikimedia Commons / https://commons.wikimedia.org/wiki/File:Sagittal_suture_-_skull_-_anterior_view02.png.) B, The superior foramina.

Median palatal suture



Lateral fossa

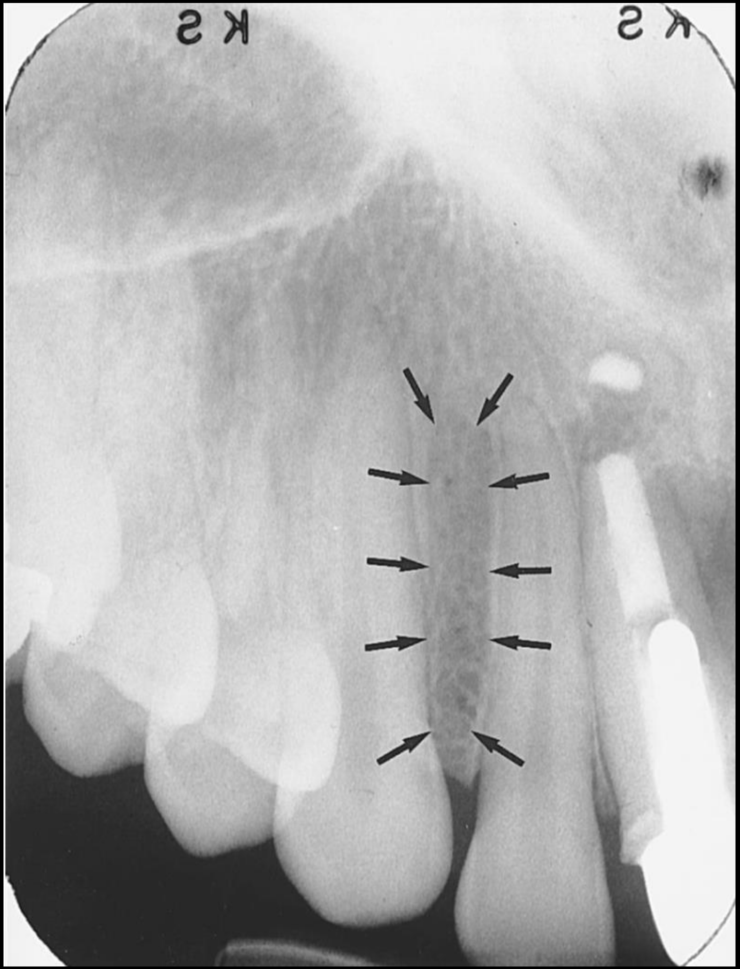


FIG 28-18 Lateral fossa. (Image created from BodyParts3D, © The Database Center for Life Science licensed under CC Attribution-Share Alike 2.1 Japan [http://creativecommons.org/licenses/by-sa/2.1/jp/legalcode] / Modified from User: Was a bee / Wikimedia Commons / https://commons.wikimedia.org/wiki/File:Coronal_suture_-_skull_-_anterior_view03.png.)

Nasal cavity

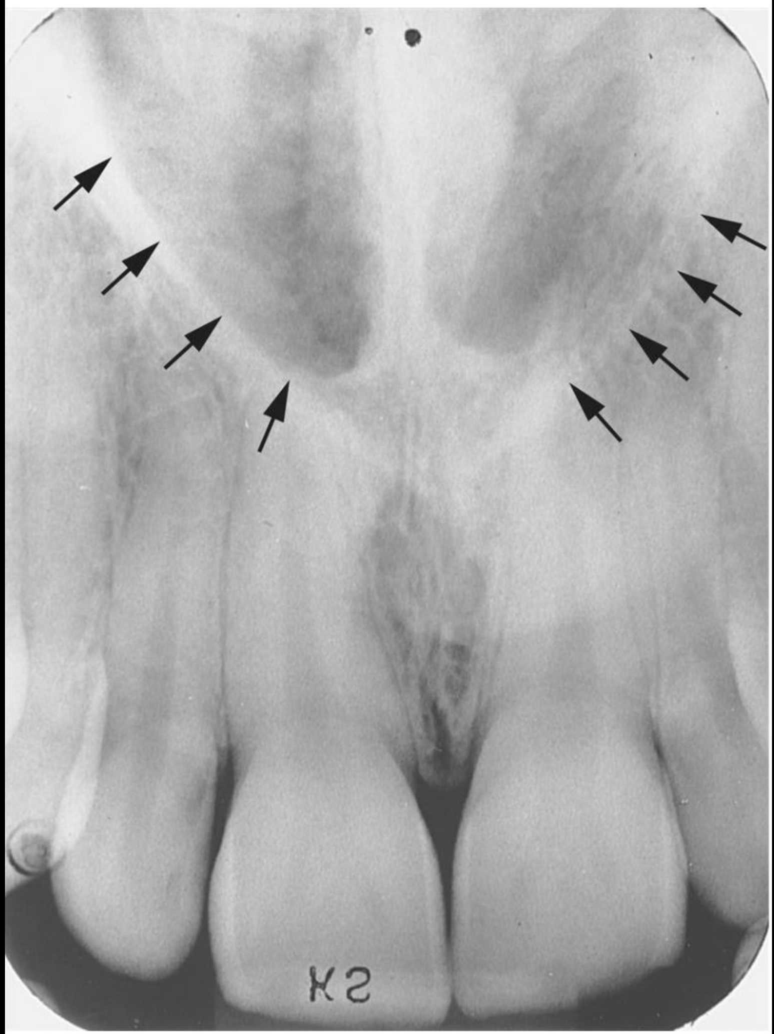
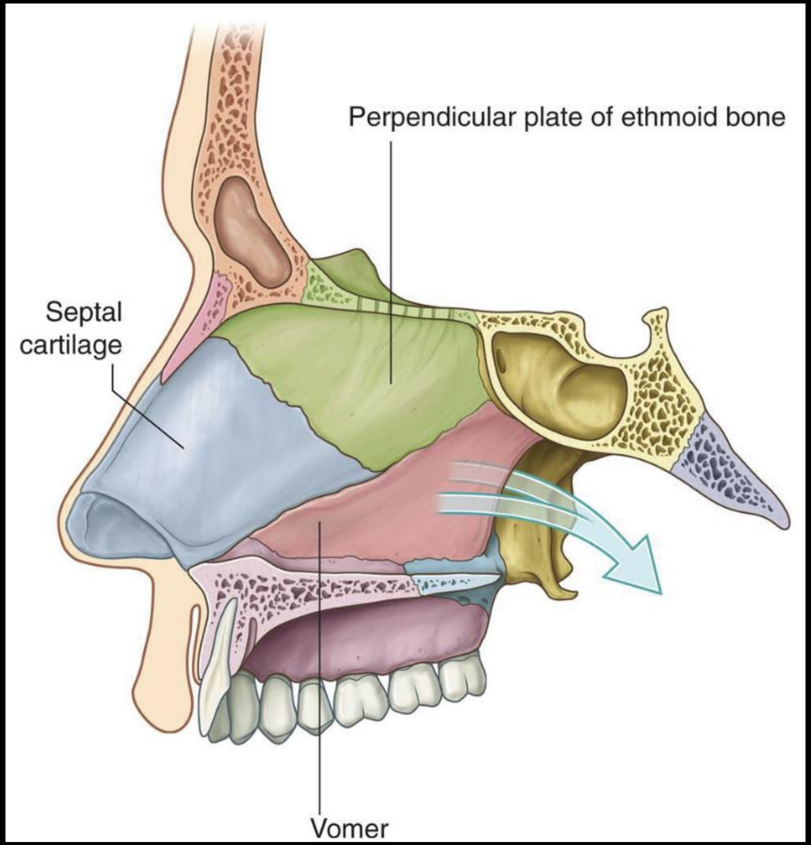
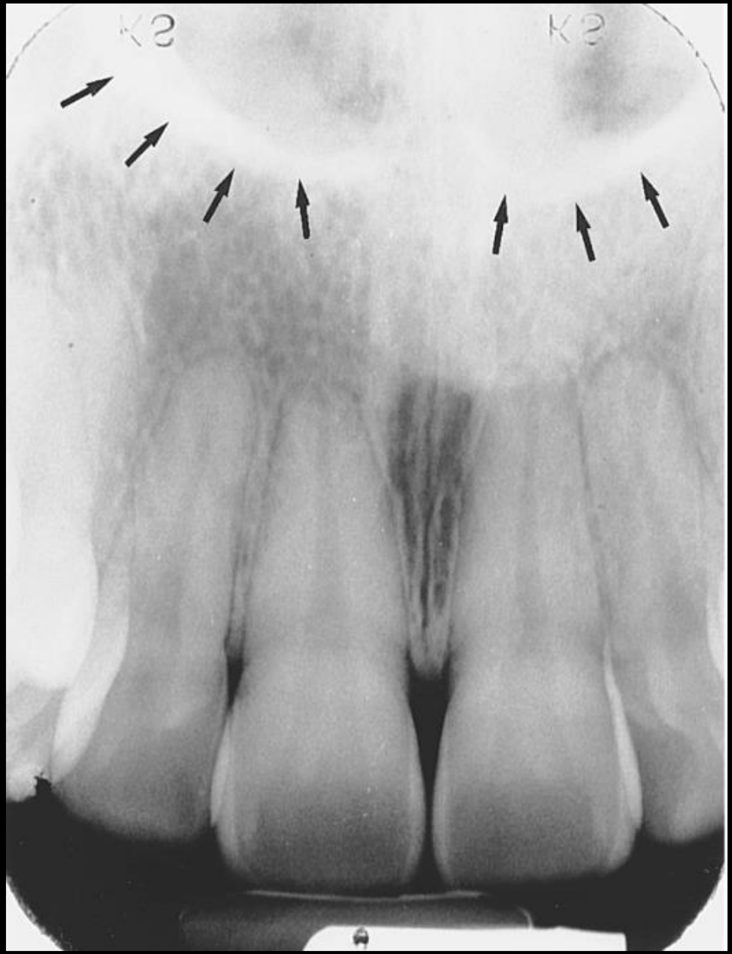


FIG 28-20 The nasal cavity appears as a large radiolucent area above the maxillary incisors. (From Haring JI, Lind LJ: Radiographic interpretation for the dental hygienist, Philadelphia, 1993, Saunders.)

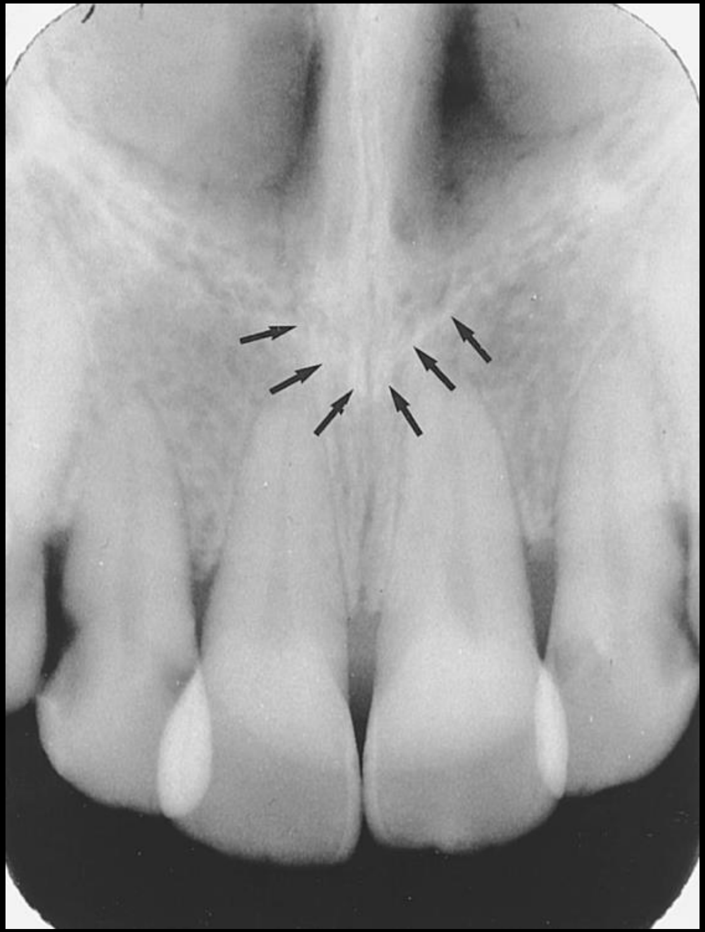
Nasal septum

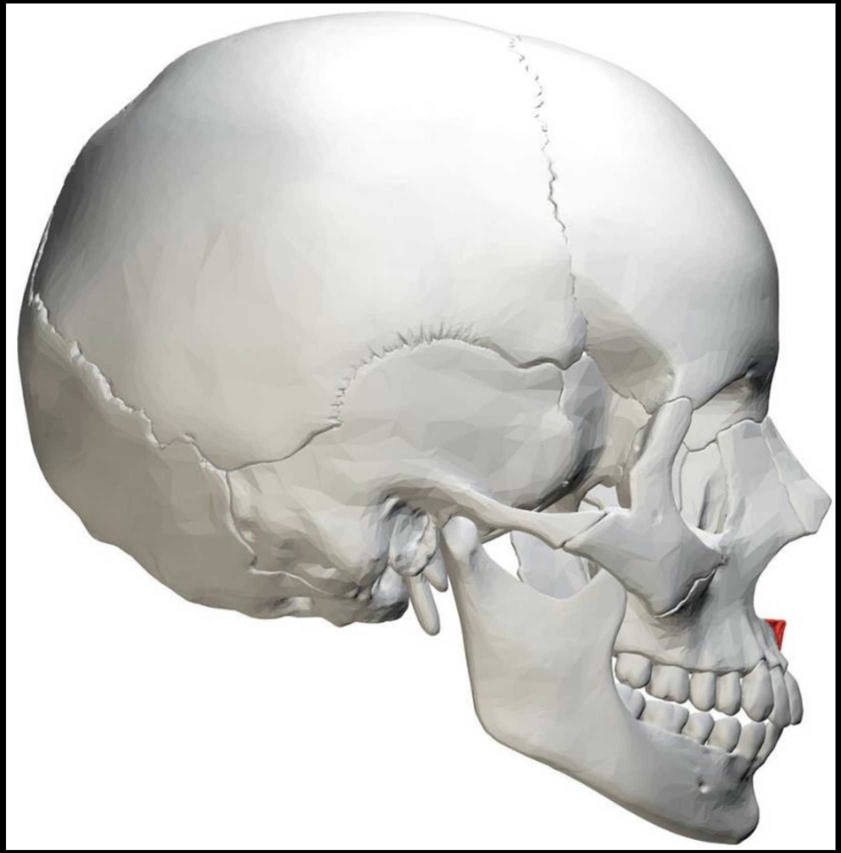


Floor of nasal cavity

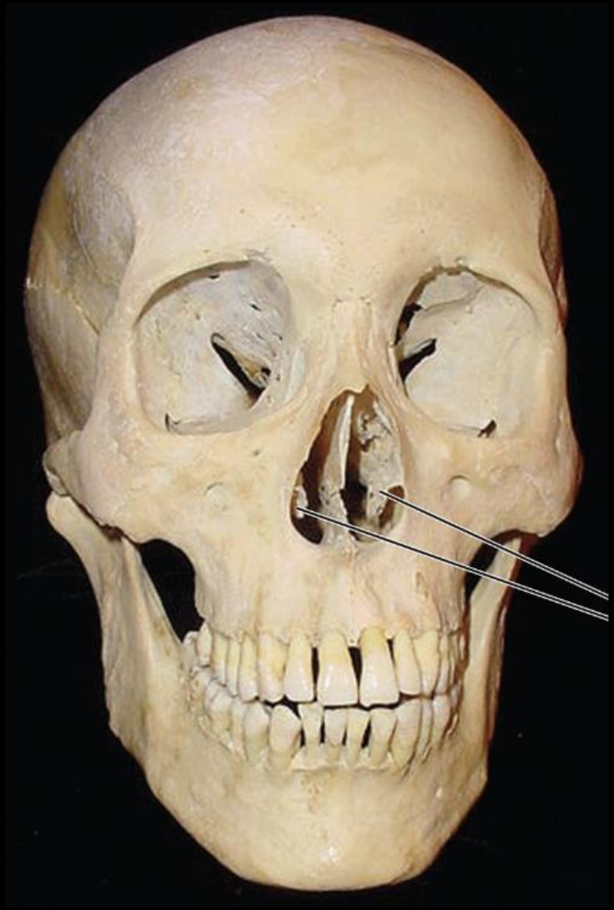


Anterior nasal spine





Inferior nasal conchae



Maxillary sinus

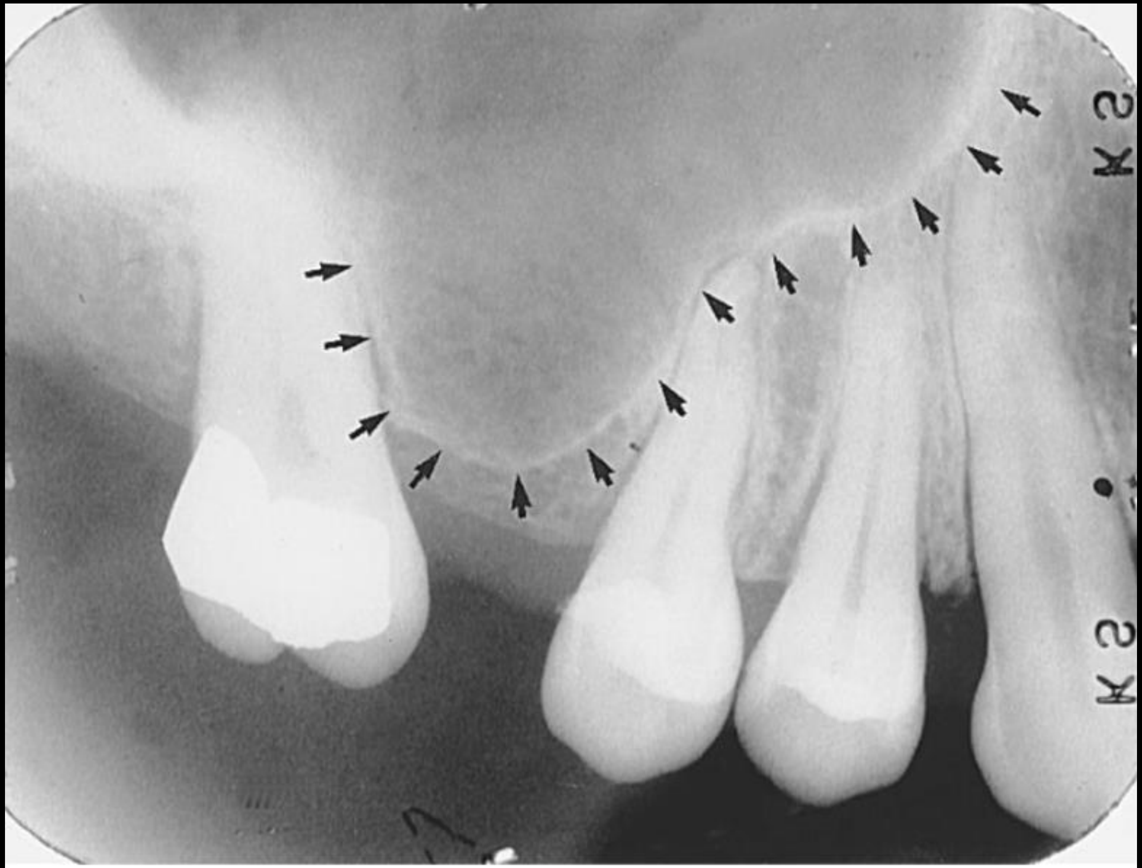


FIG 28-28 The maxillary sinus appears as a radiolucent area above the maxillary posterior teeth. (From Haring JI, Lind LJ: Radiographic interpretation for the dental hygienist, Philadelphia, 1993, Saunders.)

Nutrient canals

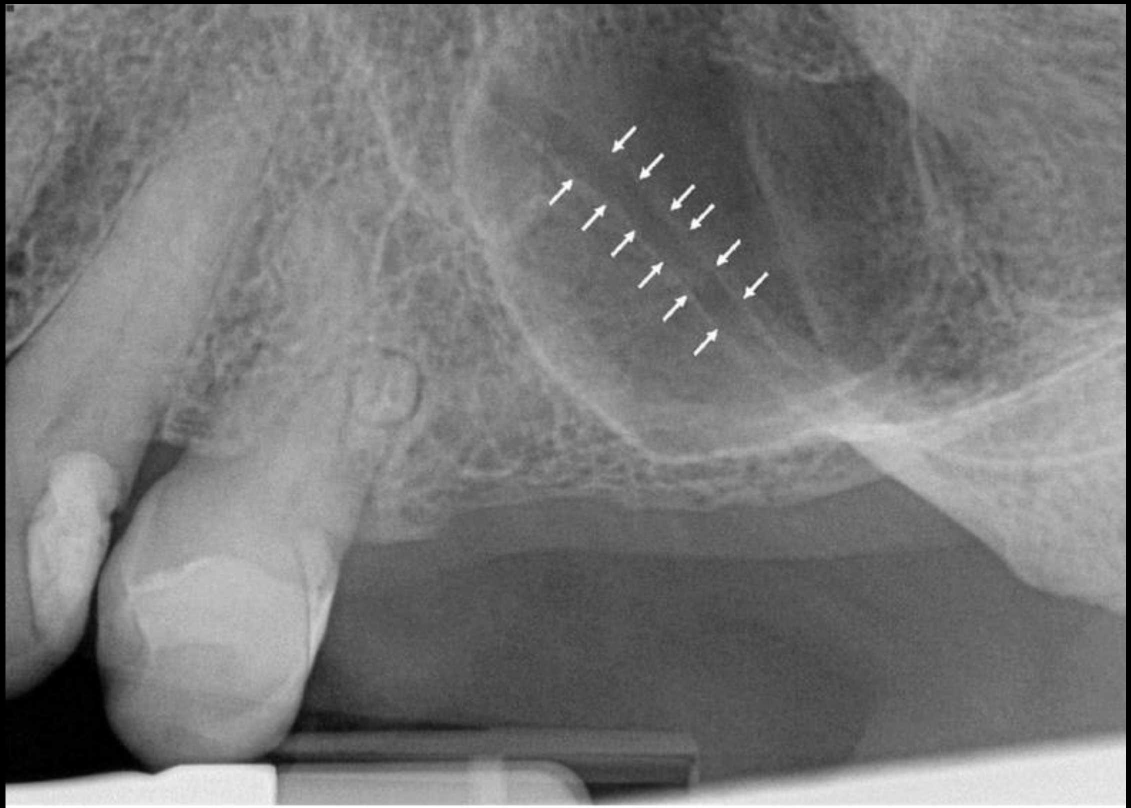


FIG 28-30 Nutrient canals appear as narrow radiolucent bands.

Inverted Y

Maxillary Tuberosity

FIG 28-32 The maxillary tuberosity appears as a radiopaque bulge distal to the third molar region. (From Haring JI, Lind LJ: Radiographic interpretation for the dental hygienist, Philadelphia, 1993, Saunders.)

Hamulus

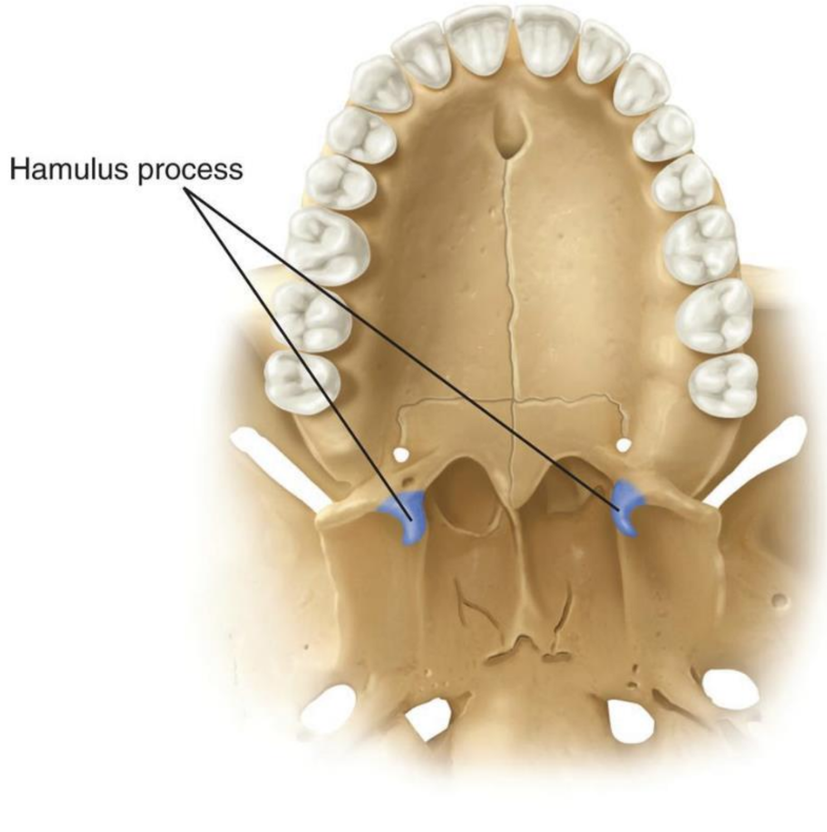


FIG 28-33 Inferior view of the hamular processes.

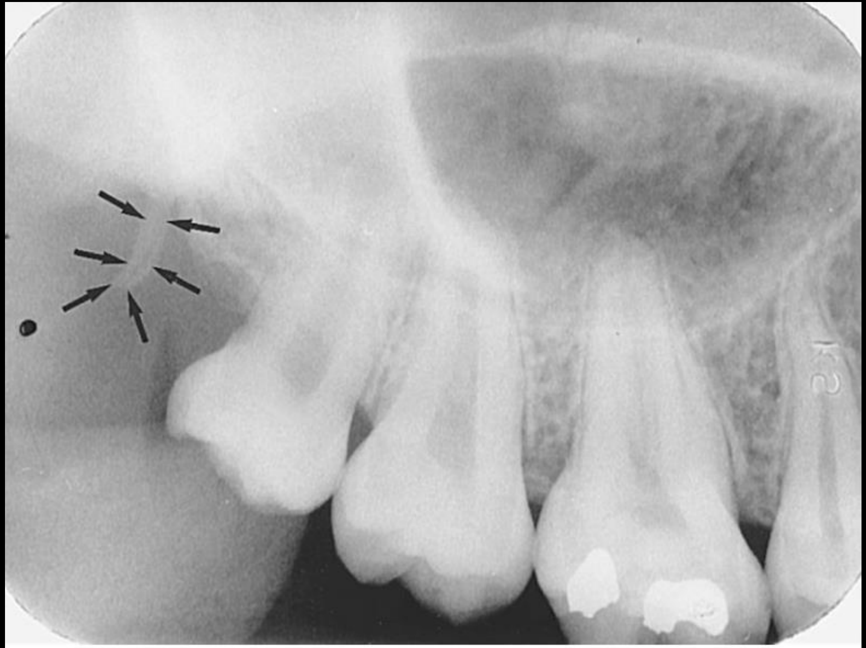


FIG 28-34 The hamulus appears as a hook-like radiopacity distal to the maxillary tuberosity area. (From Haring JI, Lind LJ: Radiographic interpretation for the dental hygienist, Philadelphia, 1993, Saunders.)

Zygomatic Process

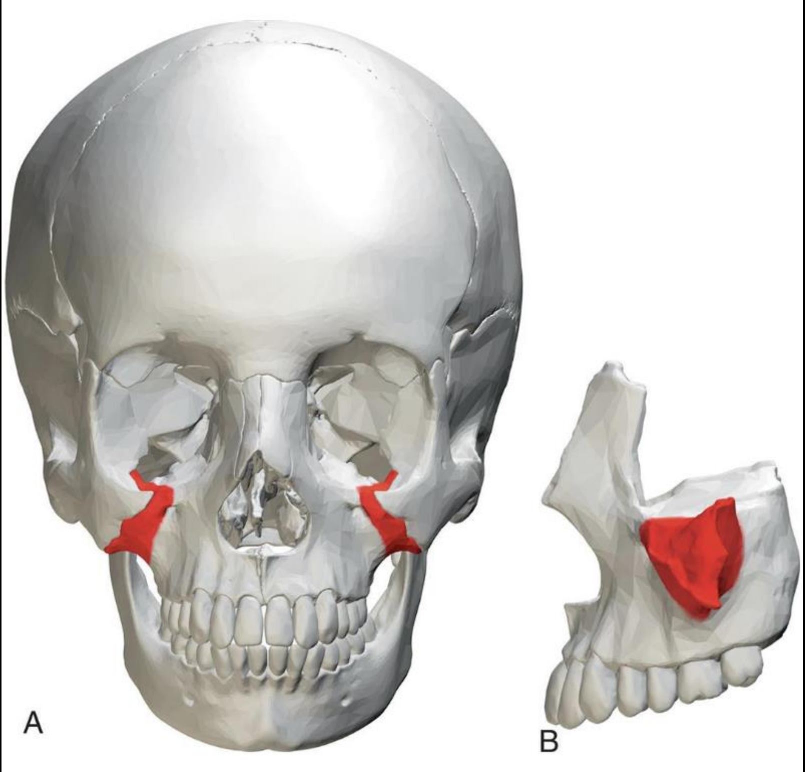
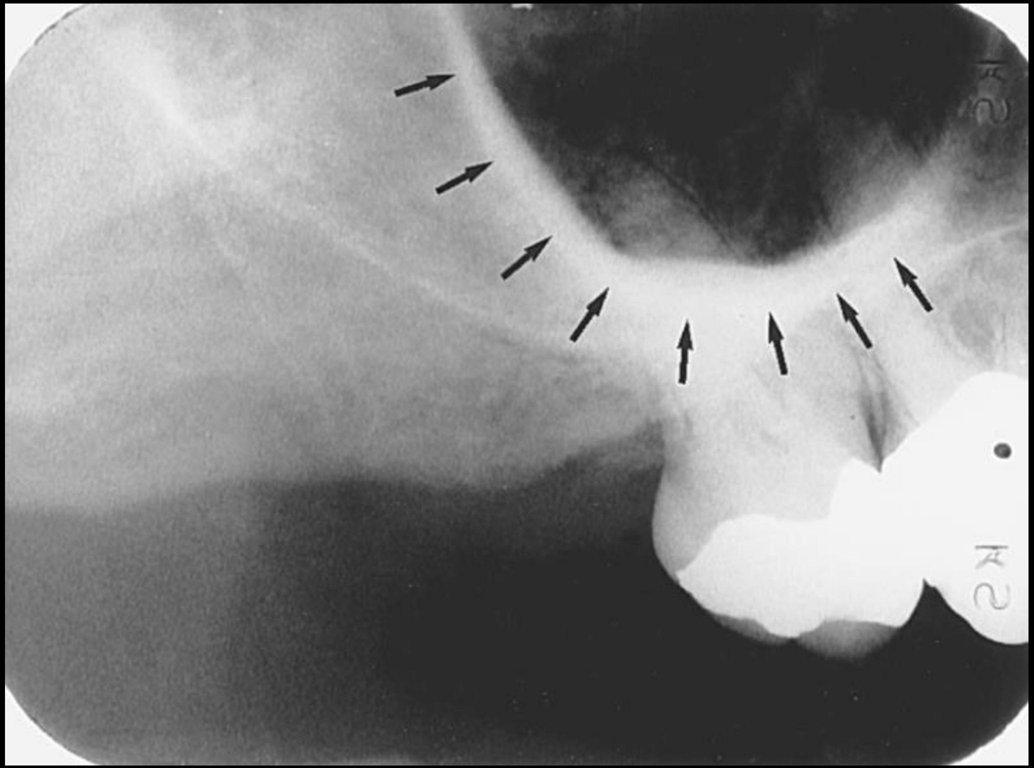


FIG 28-35 A, Zygomatic process of the maxilla, anterior view. B, Zygomatic process of the maxilla, lateral view with zygoma removed. Notice the U-shaped area of dense cortical bone. (Images created from BodyParts3D, © The Database Center for Life Science licensed under CC Attribution-Share Alike 2.1 Japan [http://creativecommons.org/licenses/by-sa/2.1/jp/legalcode] / User: Was a bee / Wikimedia Commons / A, https://commons.wikimedia.org/wiki/File:Zygomatic_process_of_maxilla_-_skull_-_anterior_view.png. B, https://commons.wikimedia.org/wiki/File:Zygomatic_process_of_maxilla_-_close_up_-_lateral_view.png.)



Zygoma

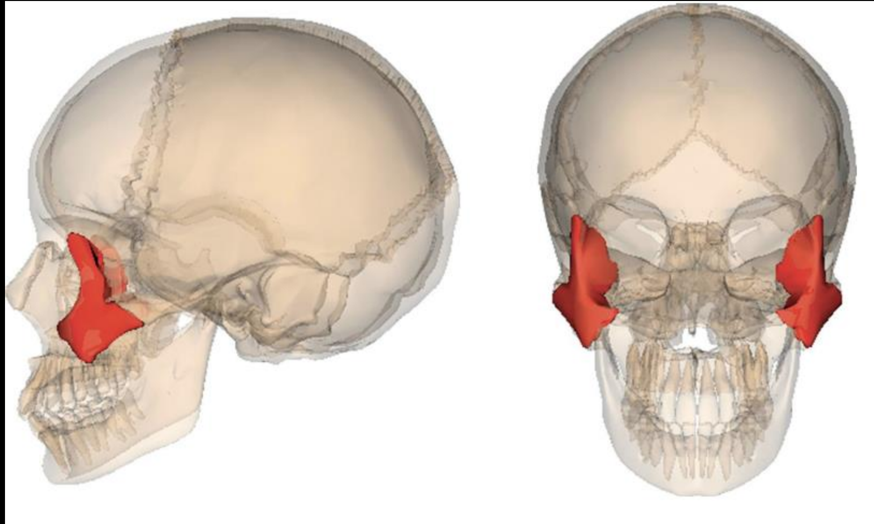


FIG 28-37 The zygoma, lateral and anterior views. (Images created from BodyParts3D, © The Database Center for Life Science licensed under CC Attribution-Share Alike 2.1 Japan [http://creativecommons.org/licenses/by-sa/2.1/jp/legalcode] / User: Was a bee / Wikimedia Commons / https://commons.wikimedia.org/wiki/File:Zygomatic_bone.png.)

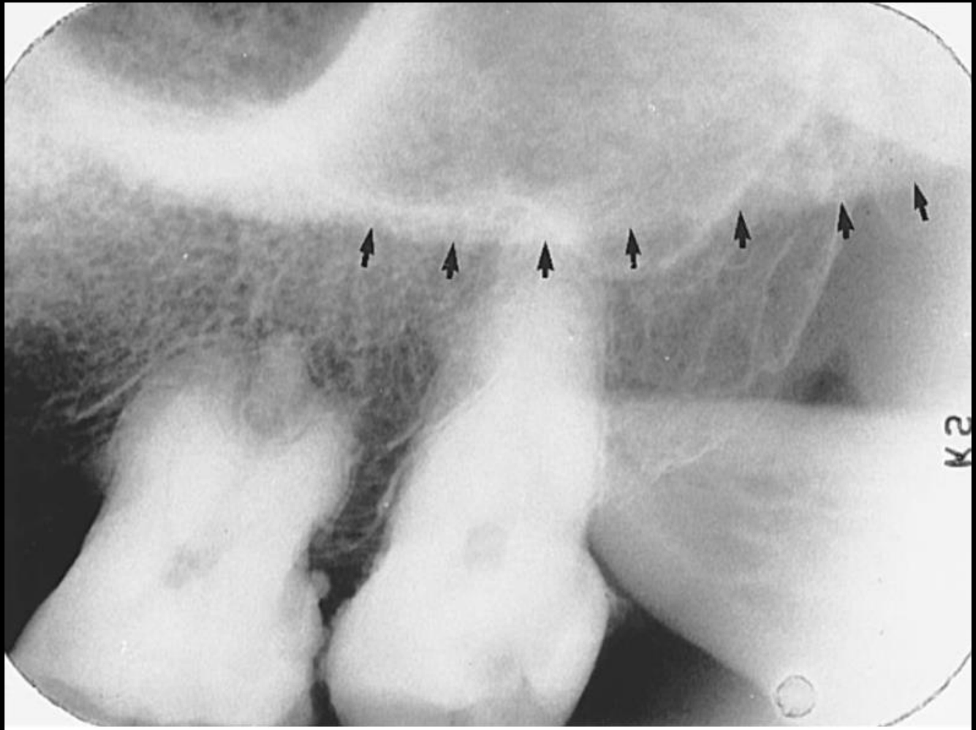
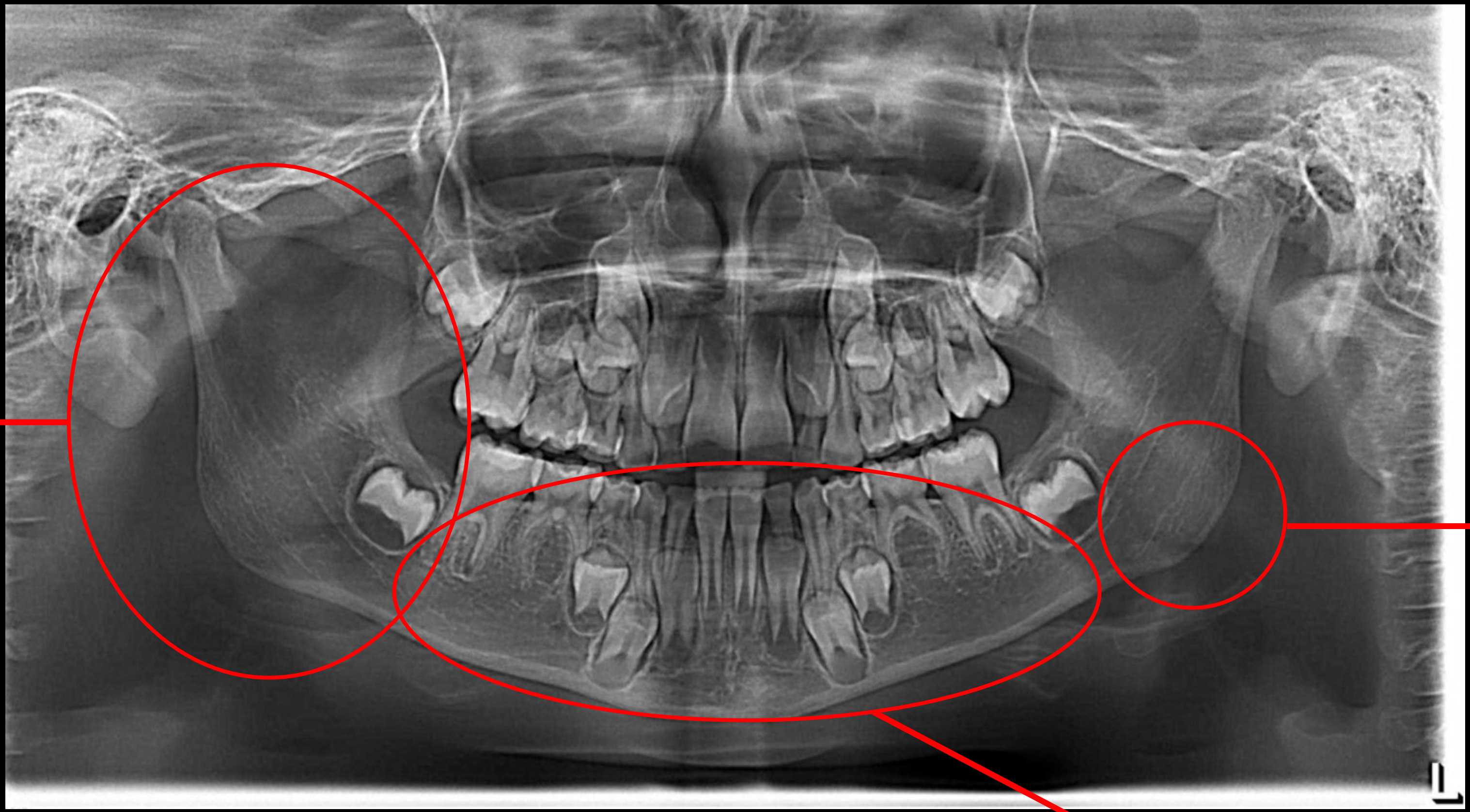


FIG 28-38 The zygoma appears as a diffuse radiopaque band that extends distally from the zygomatic process of the maxilla. (From Haring JI, Lind LJ: Radiographic interpretation for the dental hygienist, Philadelphia, 1993, Saunders.)

Normal Anatomic Landmarks- Mandible

- Ramus
- Body of the mandible
- Angle of the mandible
- Alveolar process
- Genial tubercles
- Lingual foramen
- Nutrient canals
- Mental ridge
- Mental fossa

- Mental foramen
- Mandibular canal
- Mylohyoid ridge
- External oblique ridge
- Anterior border of ramus
- Submandibular fossa
- Coronoid process


Ramus
Angle of the mandible
Body of the mandible

Genial Tubercles

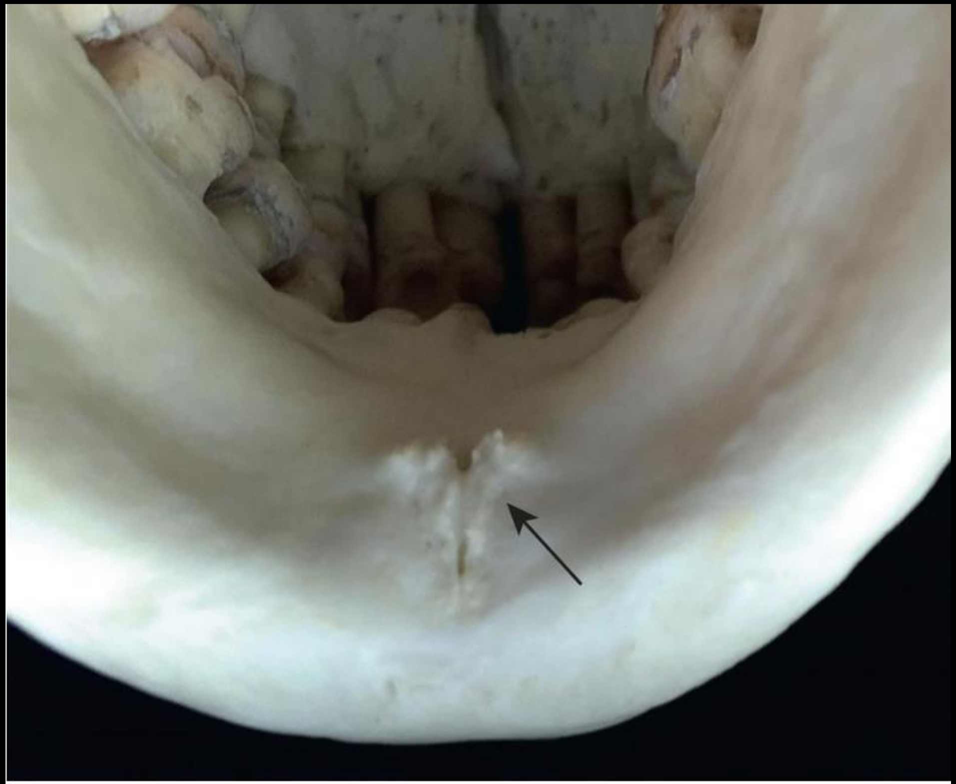


FIG 28-43 The genial tubercles.

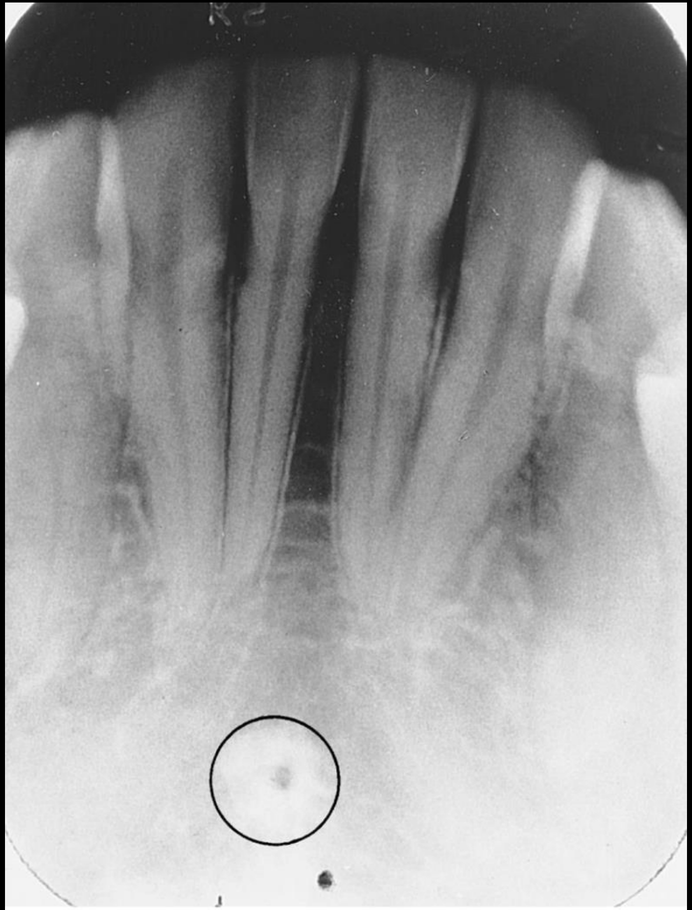
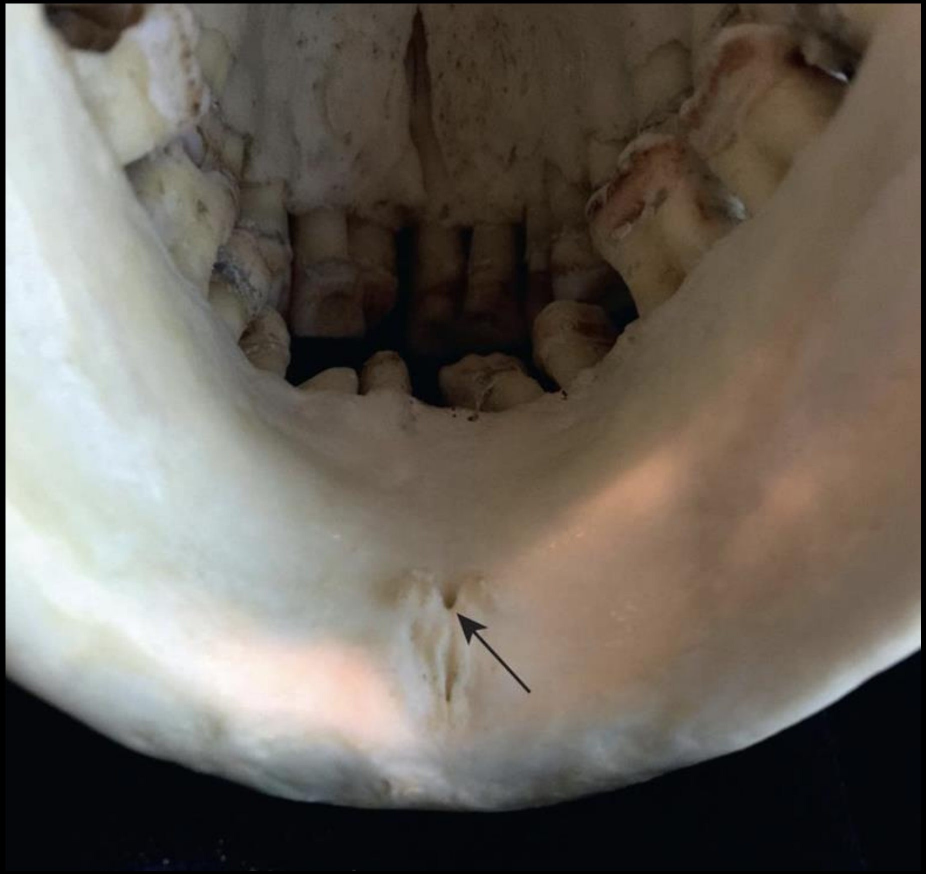
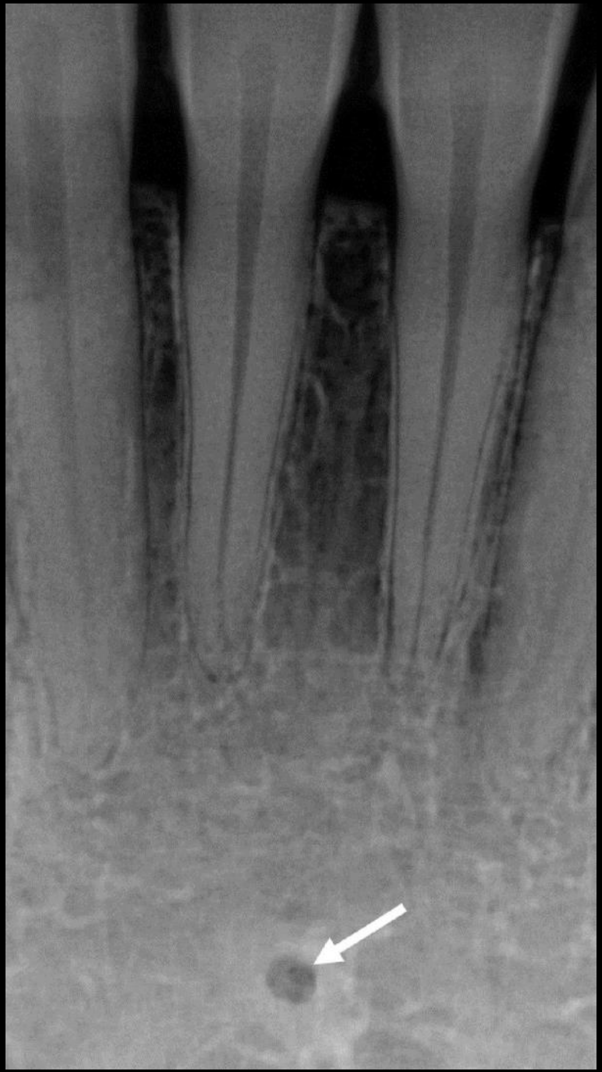


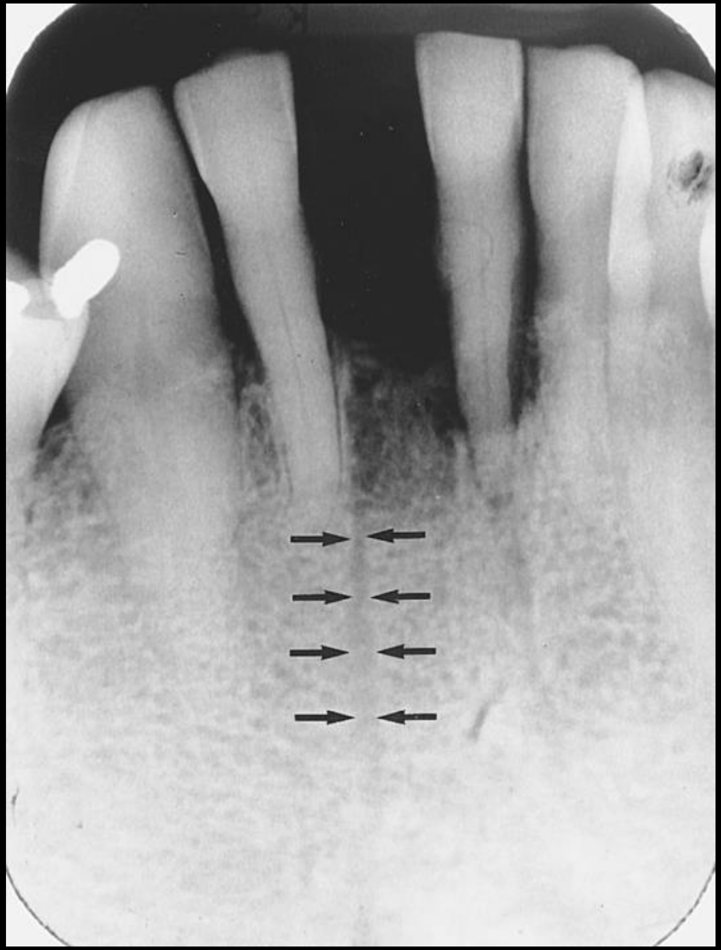
FIG 28-44 Genial tubercles appear as a ring-shaped radiopacity. (From Haring JI, Lind LJ: Radiographic interpretation for the dental hygienist, Philadelphia, 1993, Saunders.)

Lingual Foramen





Nutrient Canals



Mental Ridge & Mental Fossa

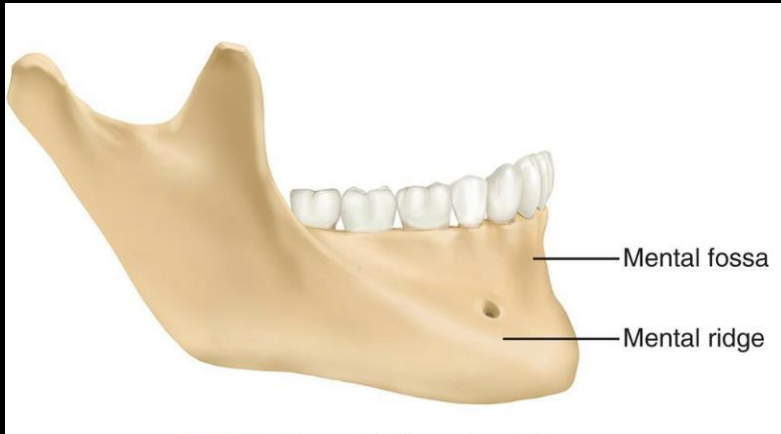
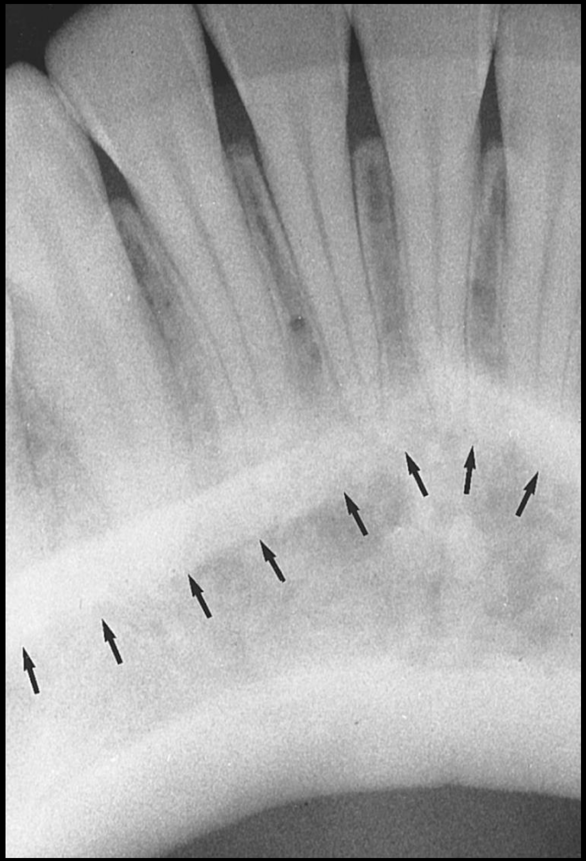
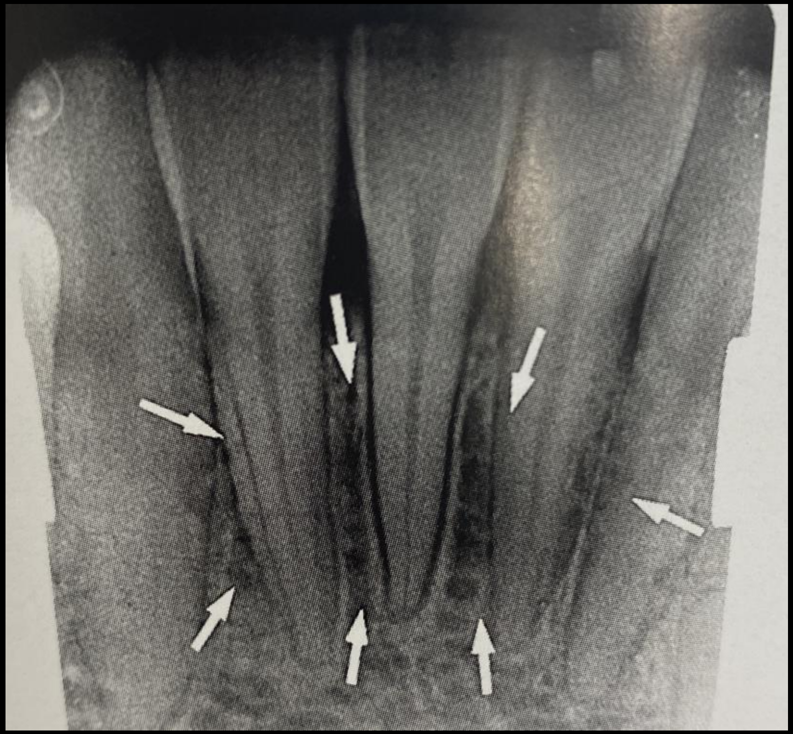


FIG 28-48 The mental ridge and mental fossa.





Mental Foramen

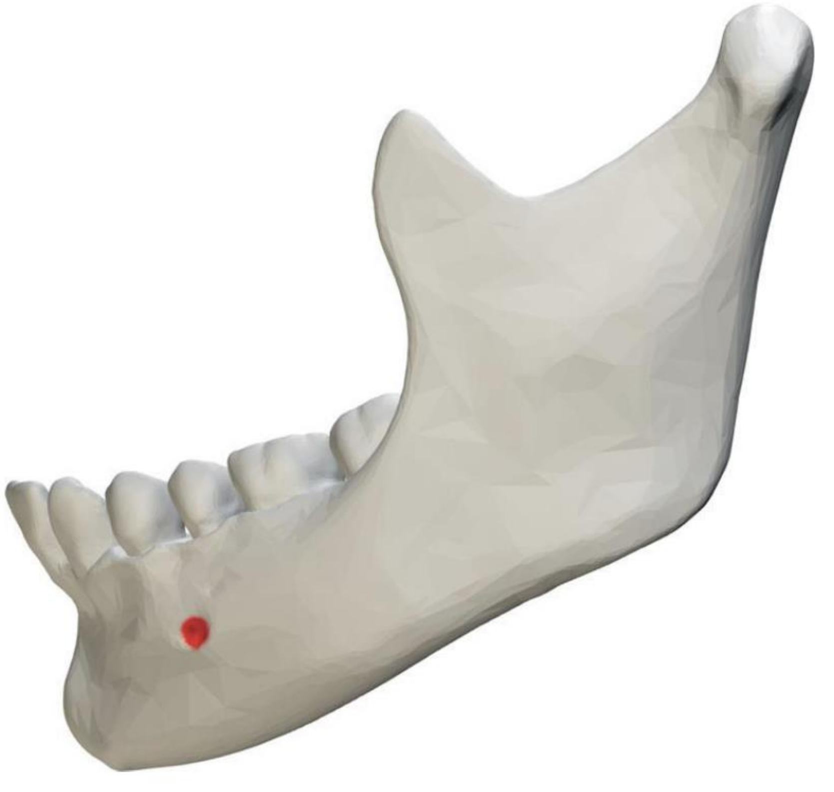


FIG 28-51 The mental foramen indicated in red. (Image created from BodyParts3D, © The Database Center for Life Science licensed under CC Attribution-Share Alike 2.1 Japan [http://

FIG 28-52 The mental foramen appears as a radiolucency in the mandibular premolar region. (From Haring JI, Lind LJ: Radiographic interpretation for the dental hygienist, Philadelphia, 1993, Saunders.)

Mandibular Canal

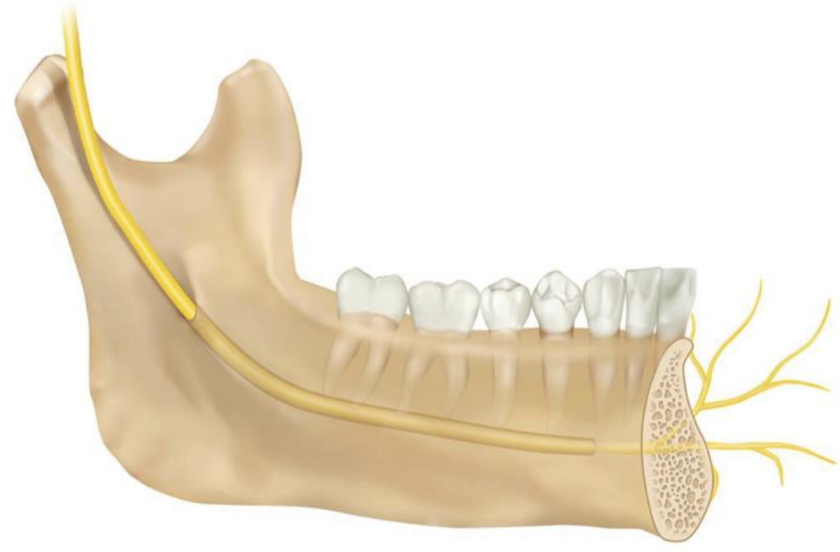


FIG 28-53 The mandibular canal.

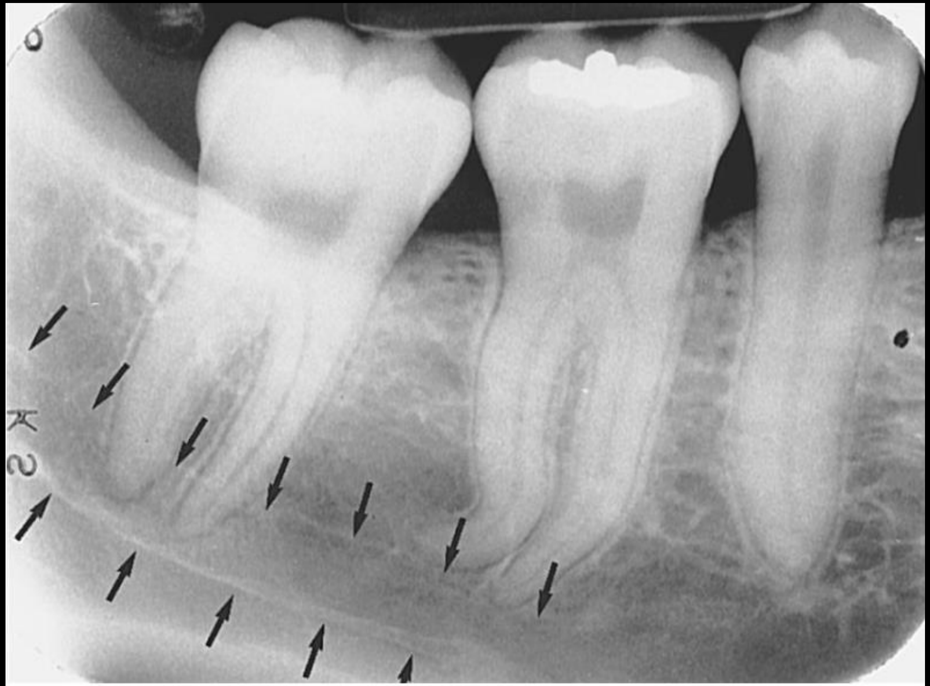


FIG 28-54 The mandibular canal appears as a radiolucent band outlined by two thin radiopaque lines. (From Haring JI, Lind LJ: Radiographic interpretation for the dental hygienist, Philadelphia, 1993, Saunders.)

Mylohyoid Ridge

FIG 28-55 A posterior view of the ramus with the mylohyoid ridge indicated in red. (Image created from BodyParts3D, © The Database Center for Life Science licensed under CC Attribution-Share Alike 2.1 Japan [http://creativecommons.org/licenses/by-sa/2.1/jp/legalcode] / User: Was a bee / Wikimedia Commons / https://commons.wikimedia.org/wiki/File:Mylohyoid_line_-_close_up_-_posterior_view.png.)

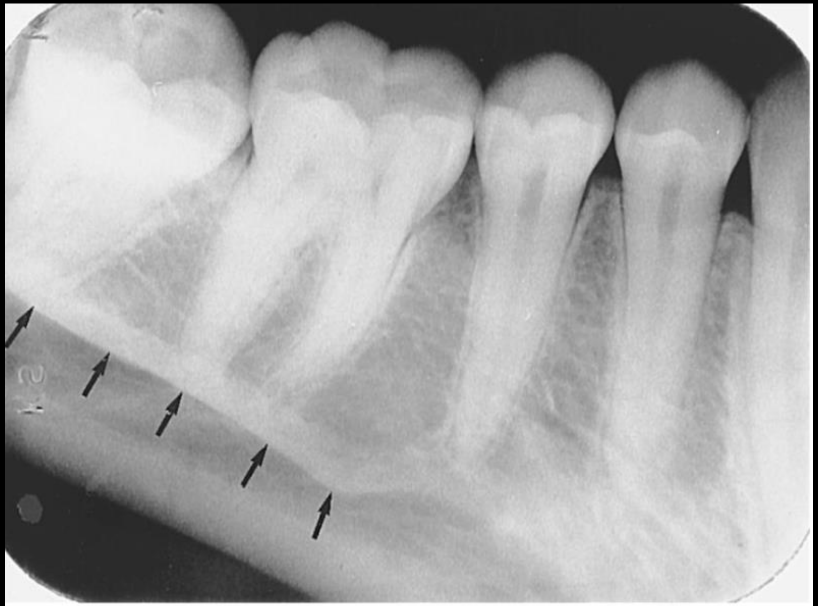


FIG 28-56 The mylohyoid ridge appears as a radiopaque band in the mandibular molar region. (From Haring JI, Lind LJ: Radiographic interpretation for the dental hygienist, Philadelphia, 1993, Saunders.)

External Oblique Ridge

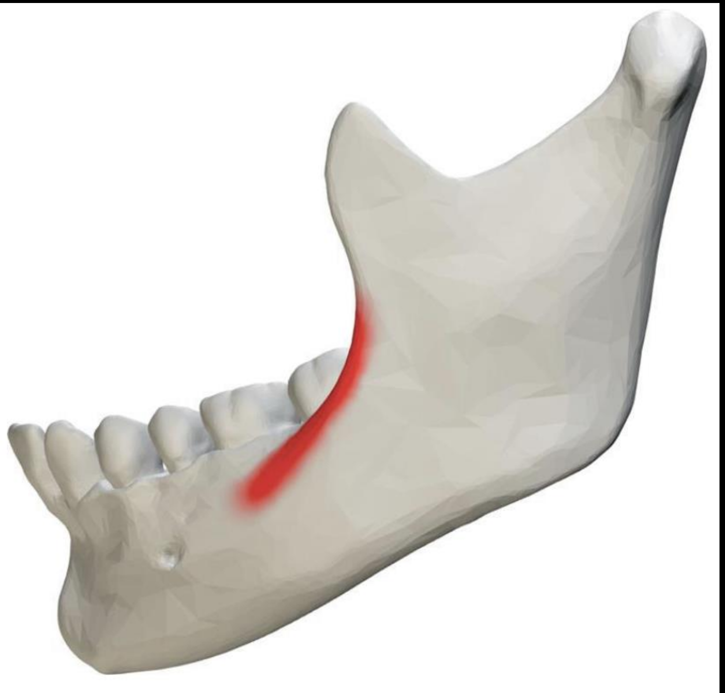


FIG 28-57 A lateral view of the mandible with the external oblique ridge indicated in red. (Image created from BodyParts3D, © The Database Center for Life Science licensed under CC Attribution-Share Alike 2.1 Japan [http://creativecommons.org/licenses/by-sa/2.1/jp/legalcode] / User: Was a bee / Wikimedia Commons / https://commons.wikimedia.org/wiki/File:External_oblique_line_of_mandible_-_close_up_-_lateral_view.png.)

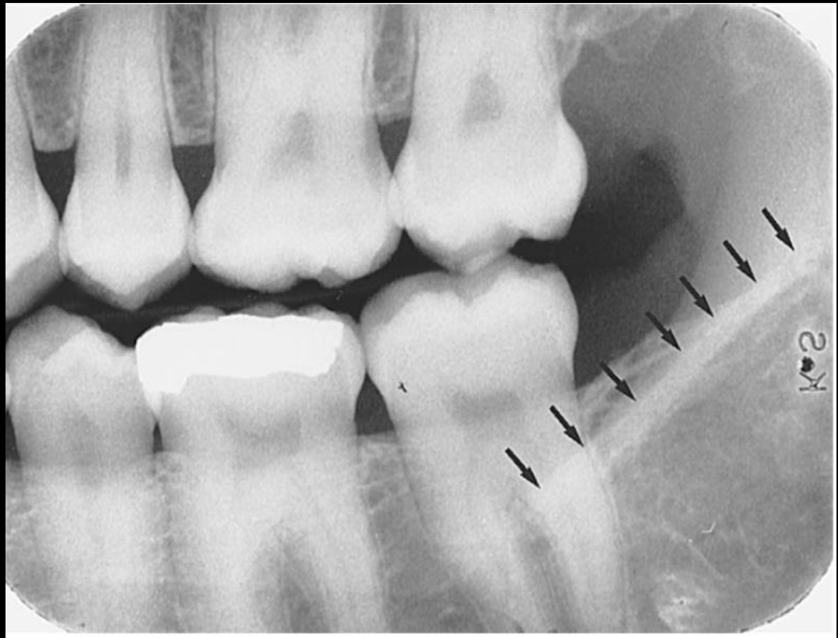


FIG 28-58 The external oblique ridge appears as a radiopaque band. (From Haring JI, Lind LJ: Radiographic interpretation for the dental hygienist, Philadelphia, 1993, Saunders.)

Anterior Border of the Ramus

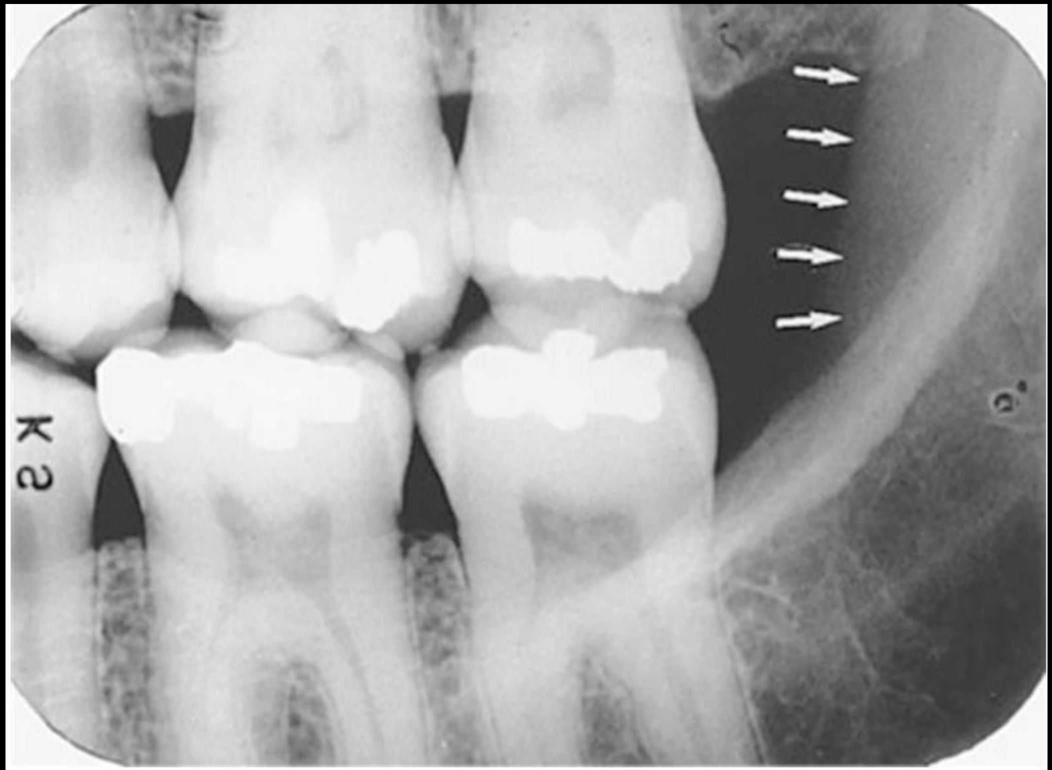


FIG 28-59 The descending ramus of the mandible appears as a vertical radiopaque band. (From Haring JI, Lind LJ: Radiographic interpretation for the dental hygienist, Philadelphia, 1993, Saunders.)

Submandibular Fossa

FIG 28-60 The submandibular fossa appears as a radiolucent area inferior to the mylohyoid ridge

Coronoid Process

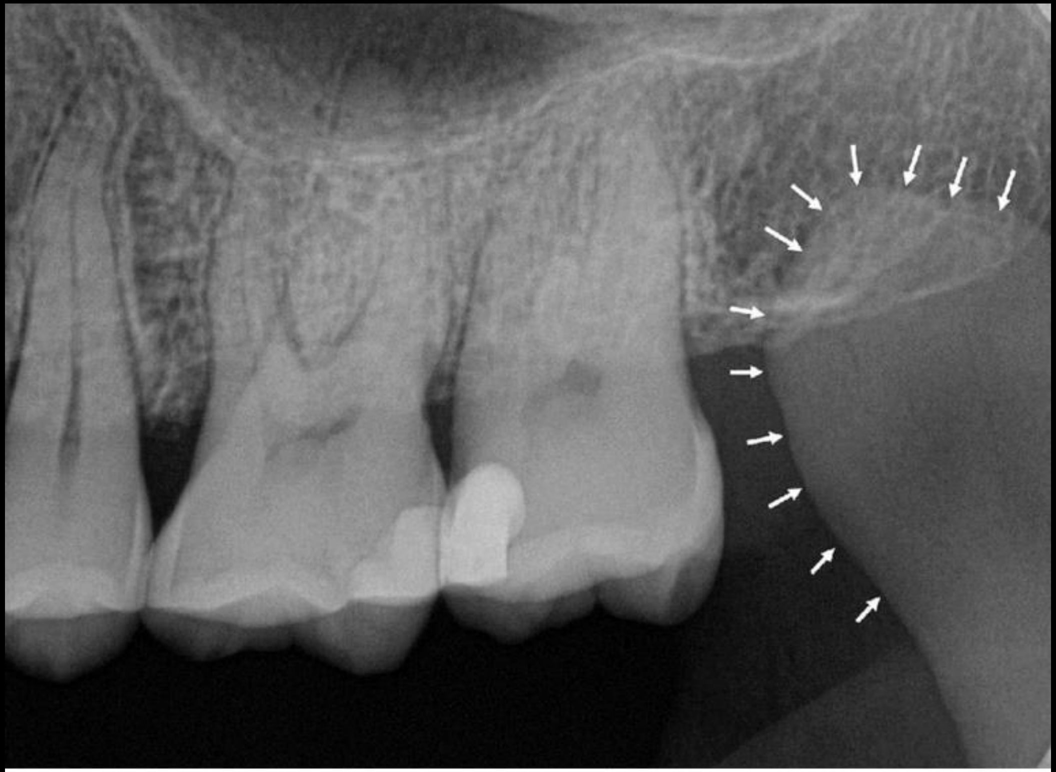


FIG 28-62 The coronoid process appears as a triangular radiopacity.

Normal Tooth Anatomy & Supporting Structures

- Enamel
- Dentin
- Dentino-Enamel Junction (DEJ)
- Pulp Cavity

- Alveolar Bone
- Alveolar crest
- Lamina dura
- Periodontal Ligament Space (PDL)

Enamel, Dentin, DEJ, Pulp

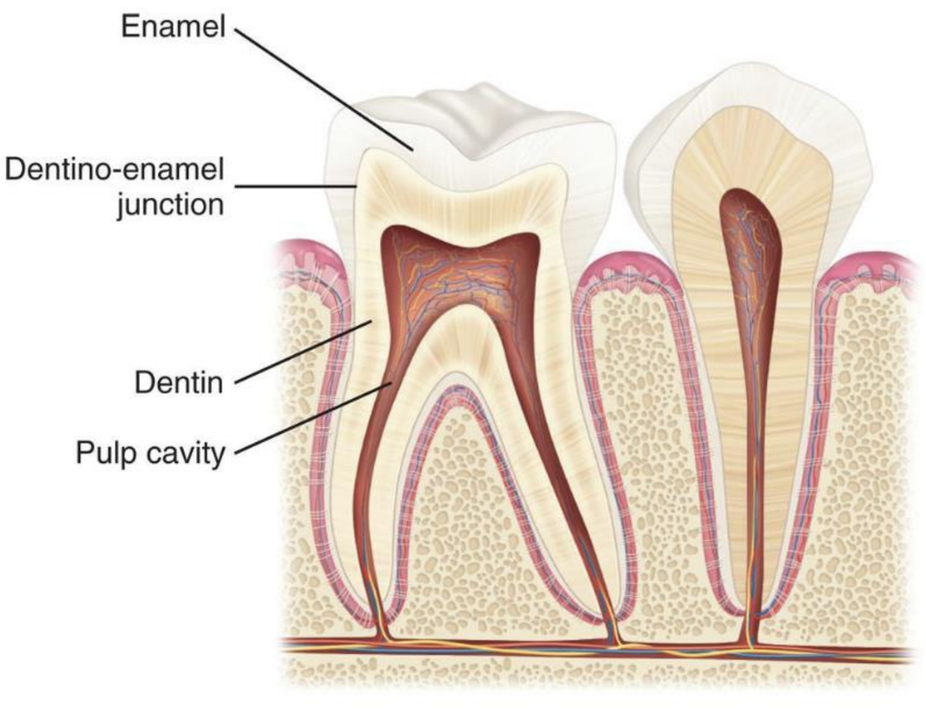


FIG 28-63 Tooth structures: enamel, dentin, dentino-enamel junction, and pulp cavity.

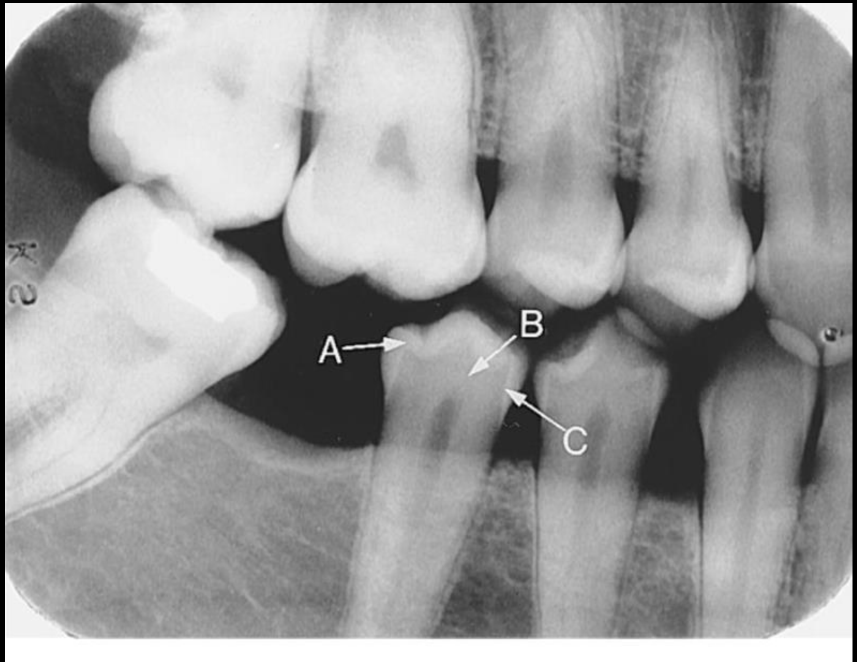


FIG 28-64 Tooth structures. **A**, enamel; **B**, dentin; **C**, dentino-enamel junction. (From Haring JI, Lind LJ: Radiographic interpretation for the dental hygienist, Philadelphia, 1993, Saunders.)

Alveolar Bone, Alveolar Crest

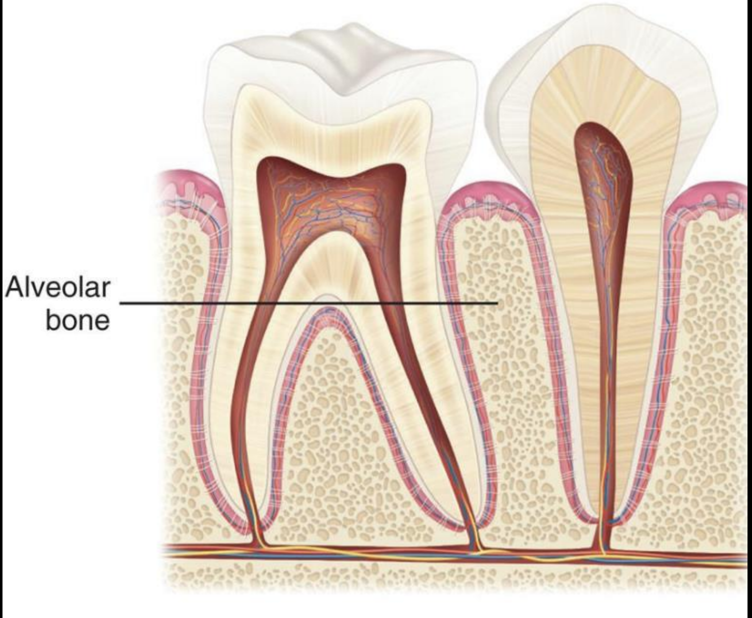


FIG 28-66 Alveolar bone.

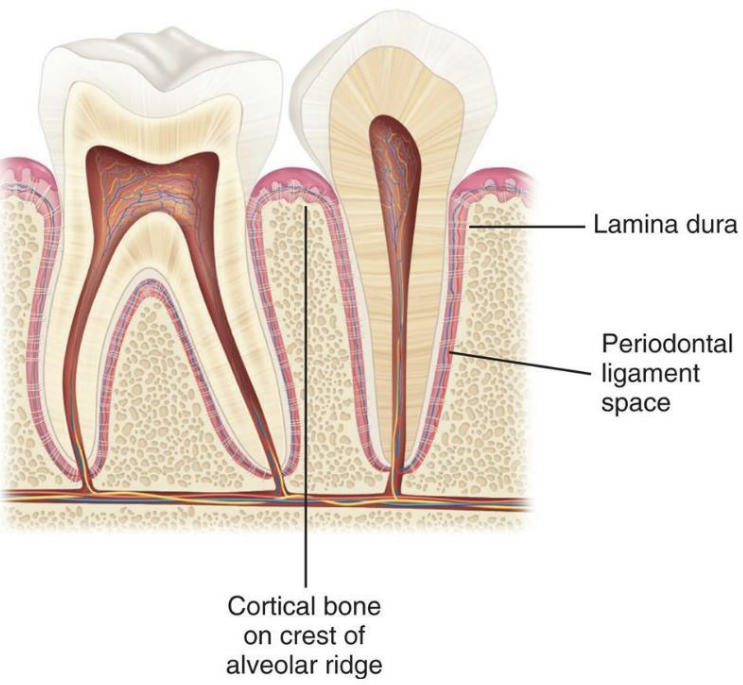


FIG 28-67 The alveolar process: lamina dura, alveolar crest, and periodontal ligament space.

FIG 28-69 The alveolar crest typically appears 1.5 to 2.0 mm below the cemento-enamel junction. (From Haring JI, Lind LJ: Radiographic interpretation for the dental hygienist, Philadelphia, 1993, Saunders.)

Lamina Dura, PDL

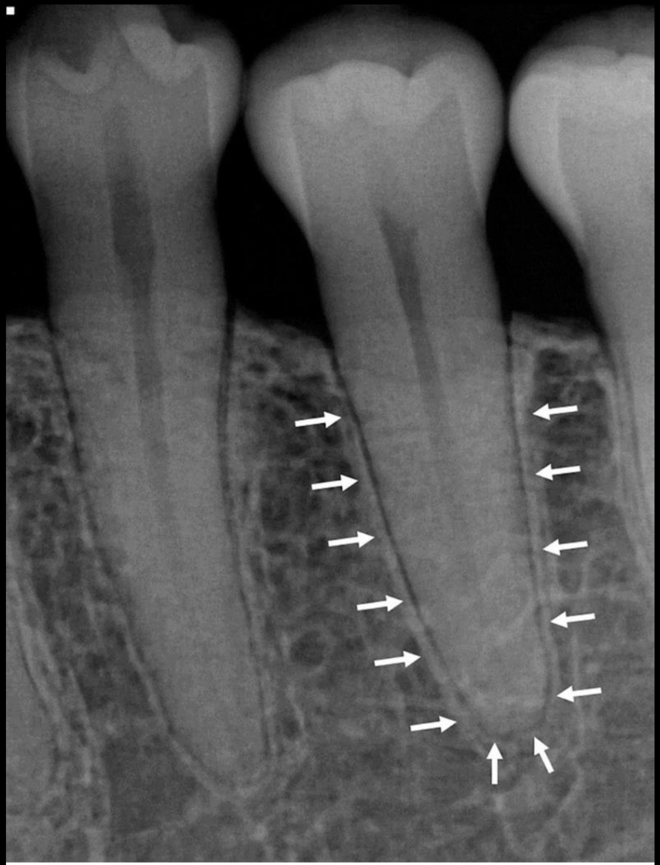


FIG 28-68 The lamina dura appears as a dense, thin radiopaque line around the root of a tooth.

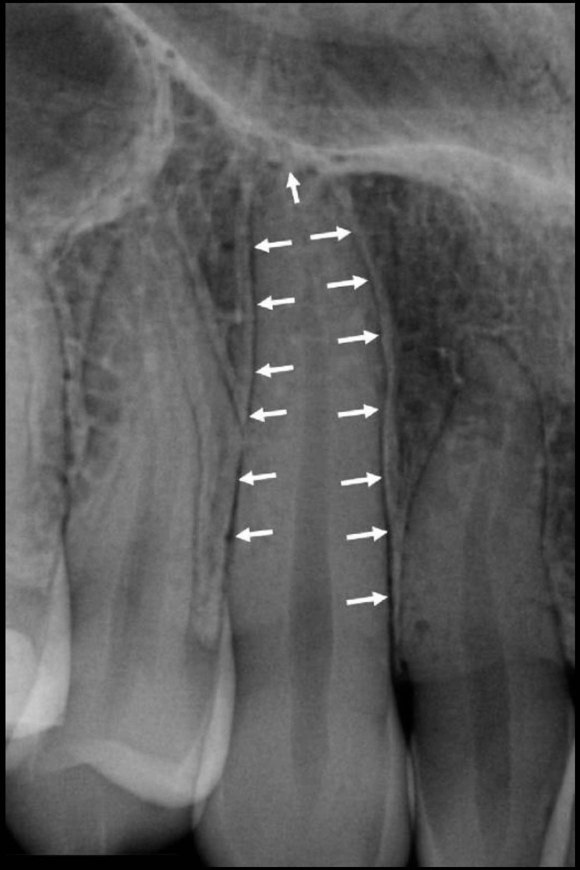


FIG 28-70 The periodontal ligament space appears as a thin radiolucent line around the root of the tooth.

Primary Dentition

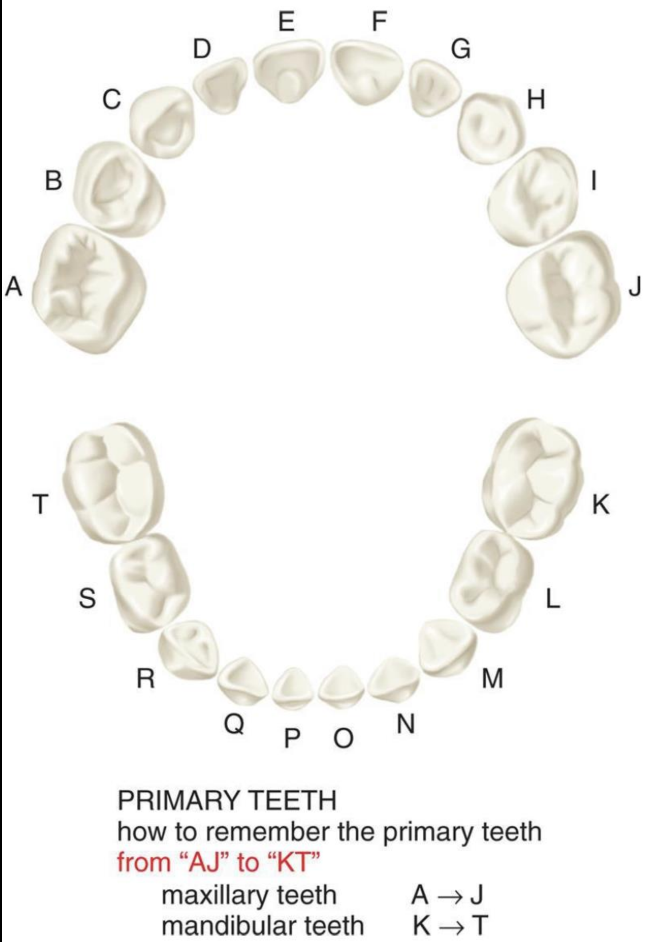
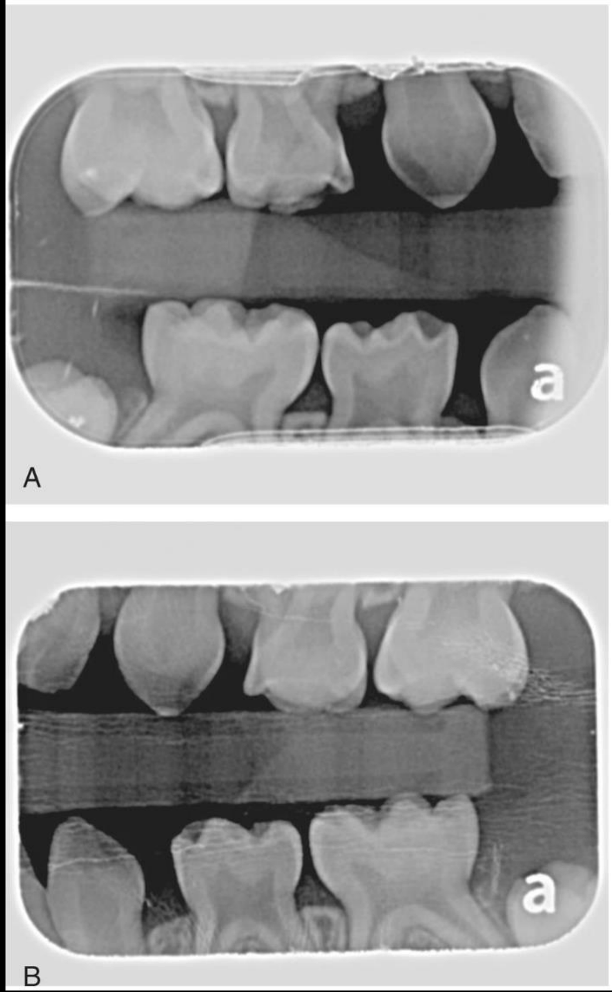


FIG 28-73 An illustration of the 20 teeth of the primary dentition labeled with the Universal Numbering System (A through T).



Mixed Dentition

Normal Edentulous Anatomy

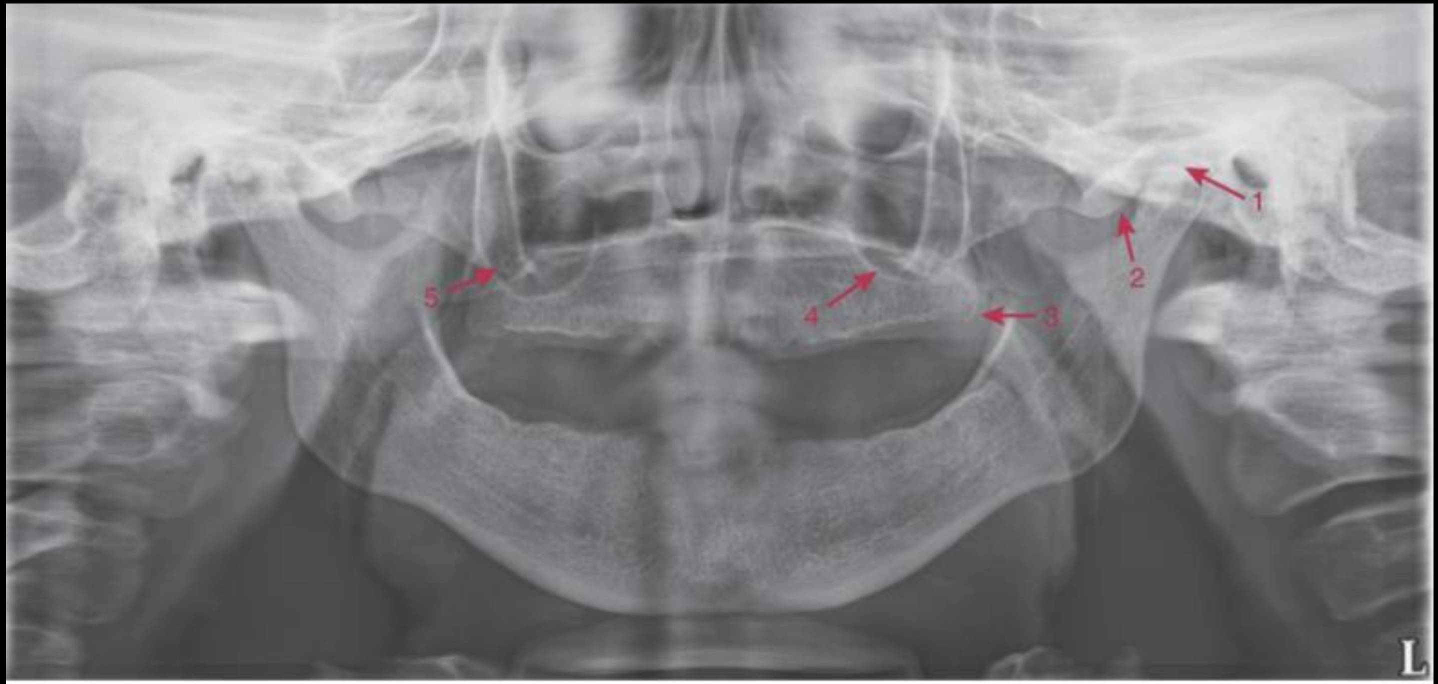


FIG 29-3 Normal anatomic landmarks of the maxilla and surrounding structures seen on a panoramic image:1, glenoid fossa; 2, articular eminence; 3, maxillary tuberosity; 4, maxillary sinus; 5, zygoma.

Identifying Restorations & Foreign Materials

- Metallic vs Non-Metallic Restorations
- Post & Core Restorations
- Porcelain Restorations
- PFM
- Composite Restorations

- Endodontic Materials
- Prosthodontic Materials
- Orthodontic Materials
- Oral Surgery Materials
- Objects: jewelry, eyeglasses

Metallic Restorations: Amalgam

FIG 32-5 Three multi-surface amalgam restorations. Note the irregular outlines.

FIG 32-6 Amalgam overhang.

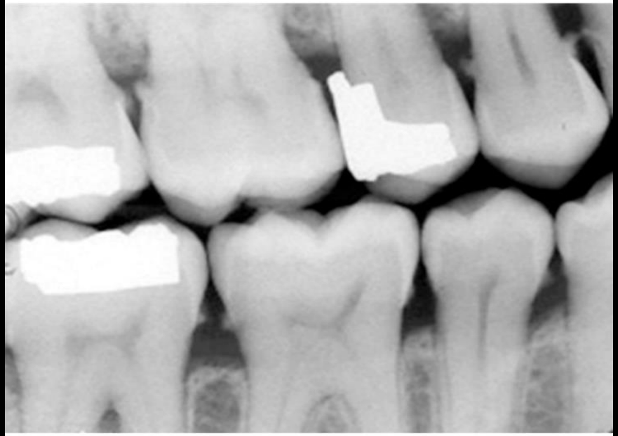


FIG 32-7 Amalgam overhang seen on the maxillary second premolar.

Metallic Restorations: Amalgam (cont.)

FIG 32-10 Amalgam tattoo seen by the bluish pigmentation of the gingiva. (From Ibsen OAC, Phelan JA: Oral pathology for the dental hygienist, ed 5, St. Louis, 2009, Saunders.)

Metallic Restorations: Gold

FIG 32-15 Gold onlays seen on maxillary premolars. Note the distinct outline and contours.

Pre-Fabricated

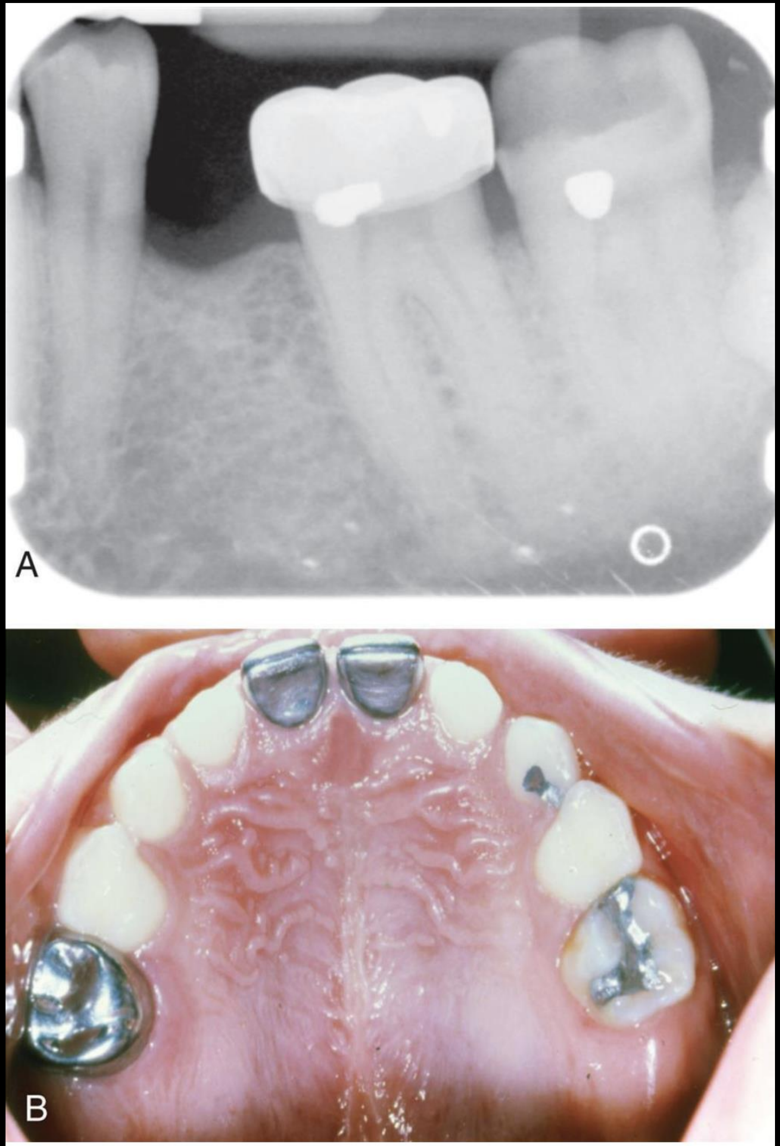


FIG 32-16 **A**, Stainless steel crown seen on the mandibular first molar. **B**, Stainless steel crown—notice the prefabricated appearance. (From Casamassimo: Pediatric dentistry: infancy through adolescence, ed 5, Saunders, 2012.)

Porcelain Restorations

FIG 32-20 All-porcelain crowns seen in the anterior maxilla. Note the outlines of the teeth preparation covered with radiopaque cement.

FIG 32-21 Porcelain bridge. (Copyright Hattanas Kumchai/iStock.com.)

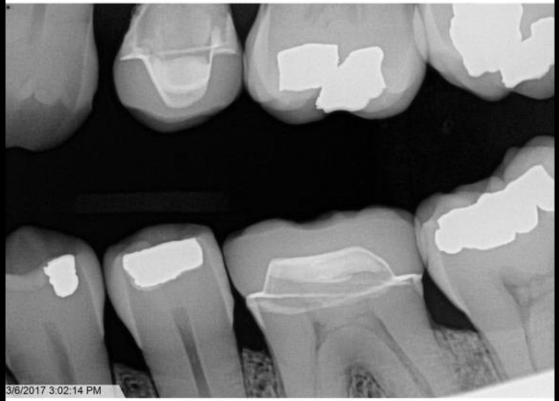


FIG 32-22 Porcelain crowns seen on the maxillary second premolar and the mandibular first molar on this bite-wing image.

Porcelain-Fused-to-Metal (PFM) Restorations

FIG 32-23 Porcelain-fused-to-metal crowns and bridges. (Courtesy W. Veneer Center, New York, NY.)

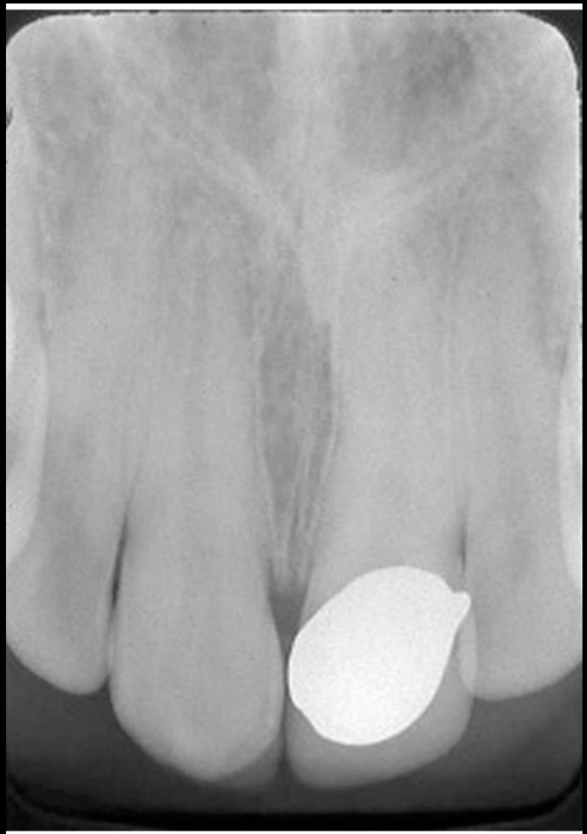


FIG 32-24 Porcelain-fused-to-metal crown seen on the maxillary central incisor.

FIG 32-25 Porcelain-fused-to-metal crown seen on the mandibular first molar.

Composite Restorations

FIG 32-26 Composite restorations seen on maxillary anterior teeth.

FIG 32-27 Composite restorations seen on mandibular anterior teeth. Note the radiolucent and radiopaque appearance of these restorations.

Endodontic Material

FIG 32-29 Metallic pin enhancing the retention capability of the composite restoration seen in this maxillary canine.

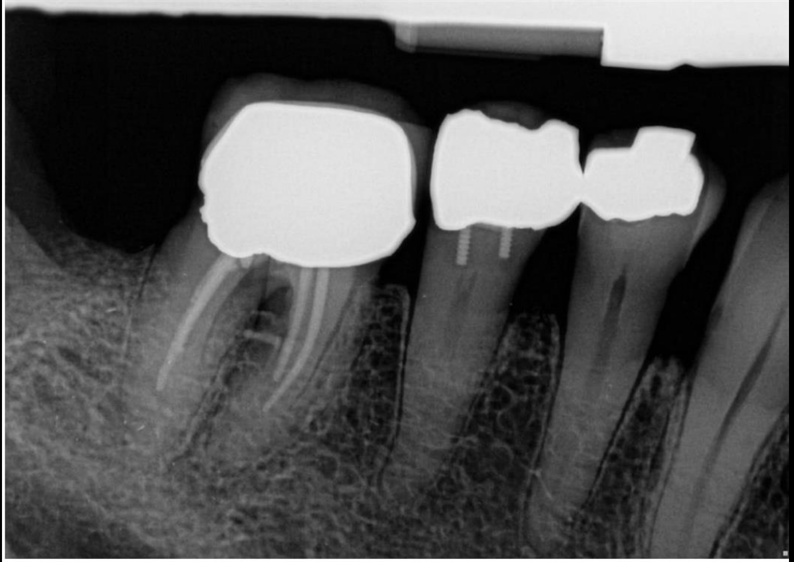
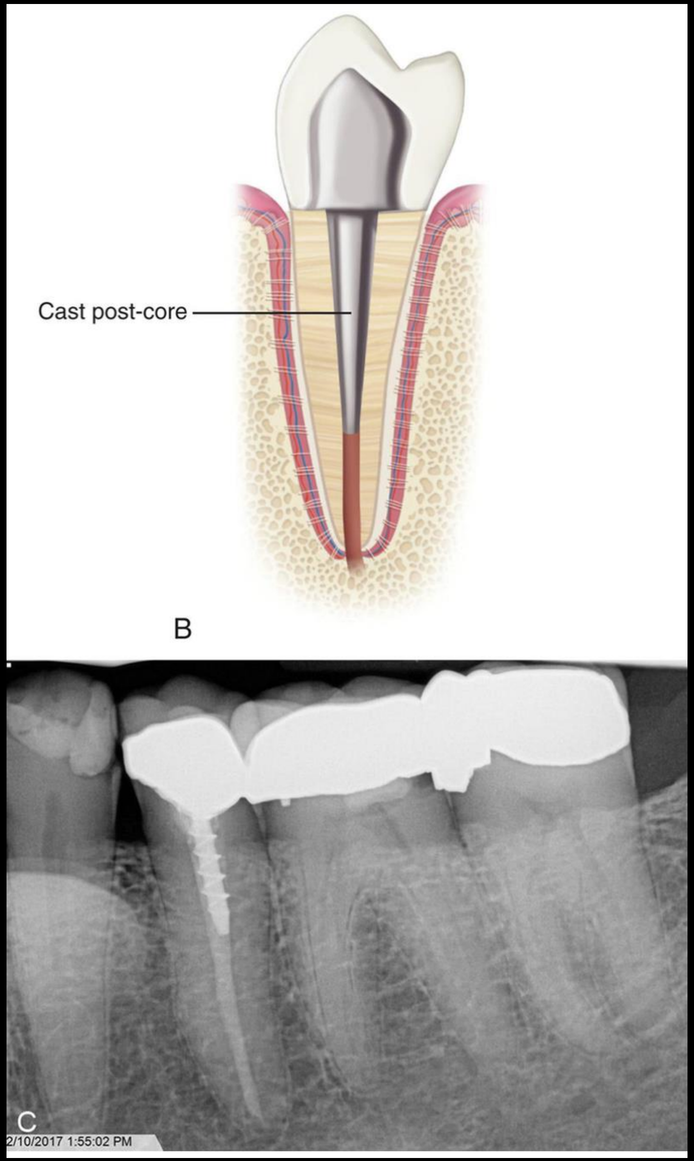


FIG 32-30 Metallic pins used to give strength to the restoration on the mandibular second premolar.

Endo Material (cont.)
Post & Core Restorations



Endo (cont.)
Gutta Percha & Silver Points

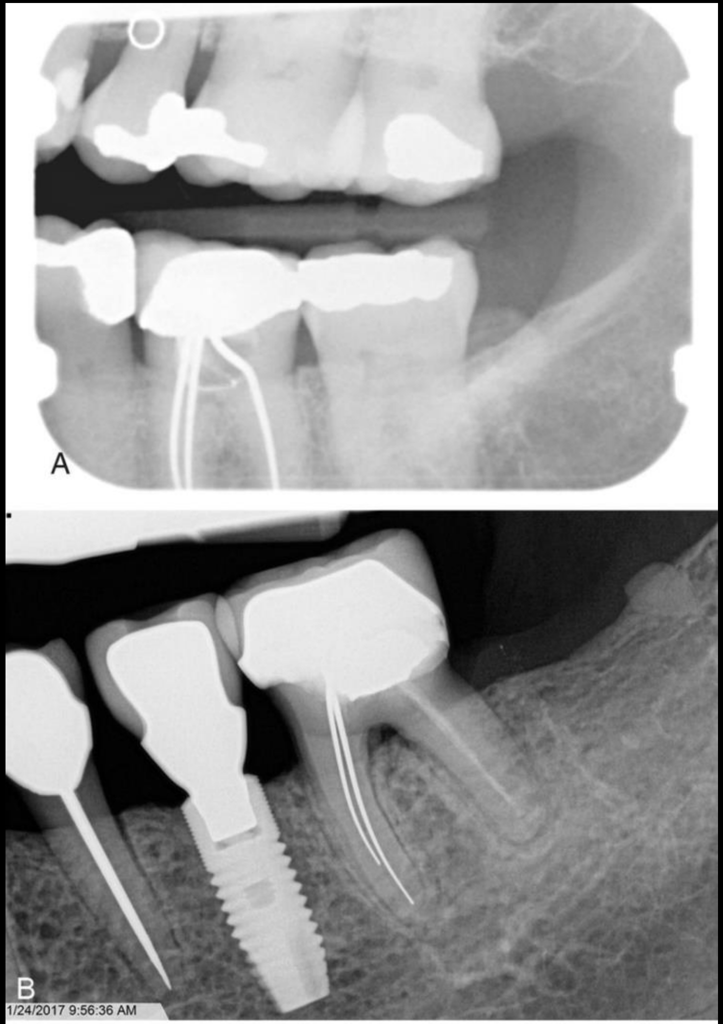


FIG 32-32 **A**, Silver point seen in the pulp canal underneath a porcelain-fused-to-metal crown on the mandibular first molar. **B**, Silver points seen in the mandibular first premolar and the mesial root of the mandibular first molar.

Prosthodontic Material: Acrylic & Porcelain

FIG 32-34 Maxillary and mandibular denture teeth appear to be "floating" in this panoramic image.

Prosth (cont)
Metal Framework

FIG 32-35 **A**, Metal framework of both partial dentures and porcelain teeth in the posterior quadrants of both arches is visible in this panoramic image. **B**, This panoramic image was exposed with the patient wearing a removable mandibular partial denture.

Orthodontic Materials

FIG 32-36 Orthodontic appliances that aid in the proper eruption of the maxillary canines.

FIG 32-37 Orthodontic bands are recognizable on a panoramic image.

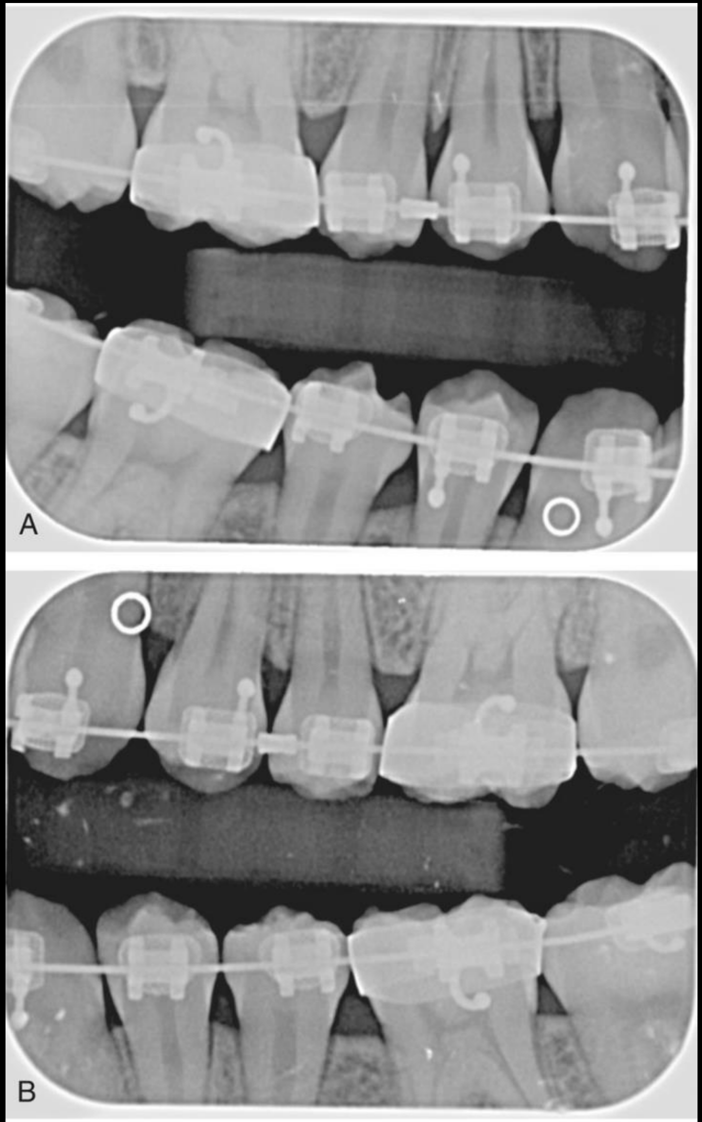


FIG 32-38 A, B, Orthodontic appliances are recognizable on bite-wing images. (Courtesy Cary Pediatric Dentistry, Cary, NC.)

Oral Surgery Materials

FIG 32-41 A dental implant is placed in the area of the maxillary lateral incisor.

FIG 32-43 Dental implants are seen in both maxillary and mandibular arches, replacing all natural teeth. (Courtesy Timothy W. Godsey, DDS, MS, Chapel Hill Periodontics and Implants, Chapel Hill, NC.)

FIG 32-46 Suture wires, screws, and stabilizing metal plates seen on a panoramic image.

FIG 32-47 Metal plate and screws seen on a panoramic image.

Foreign Materials: Earrings

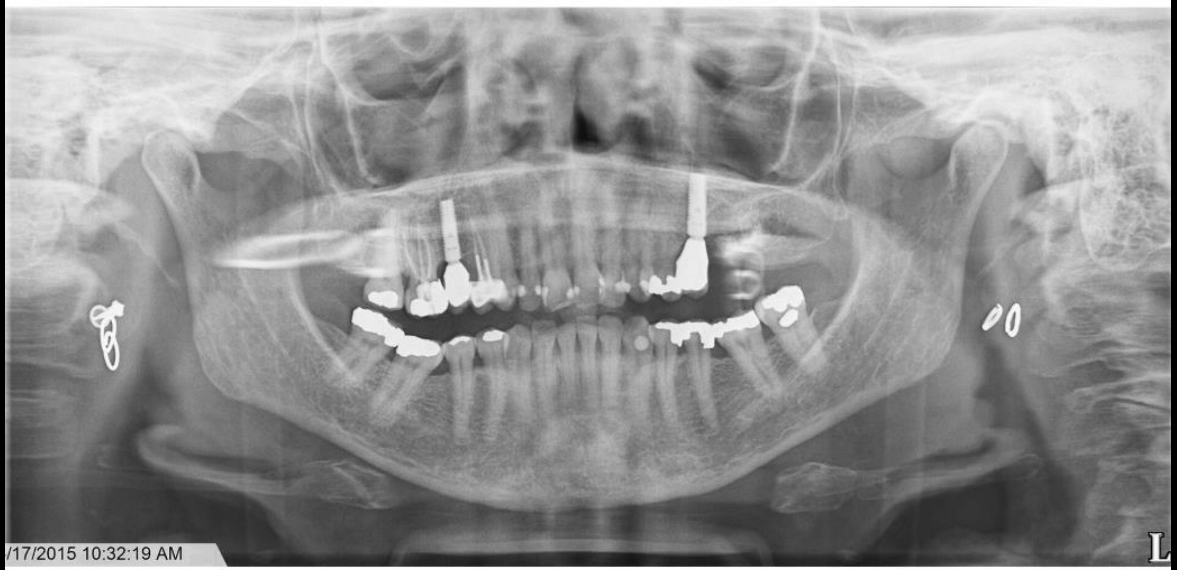


FIG 32-48 Earrings and ghost images seen on a panoramic image.

FIG 32-49 Tragus piercings and ghost images seen on a panoramic image.

Foreign Materials: Necklace

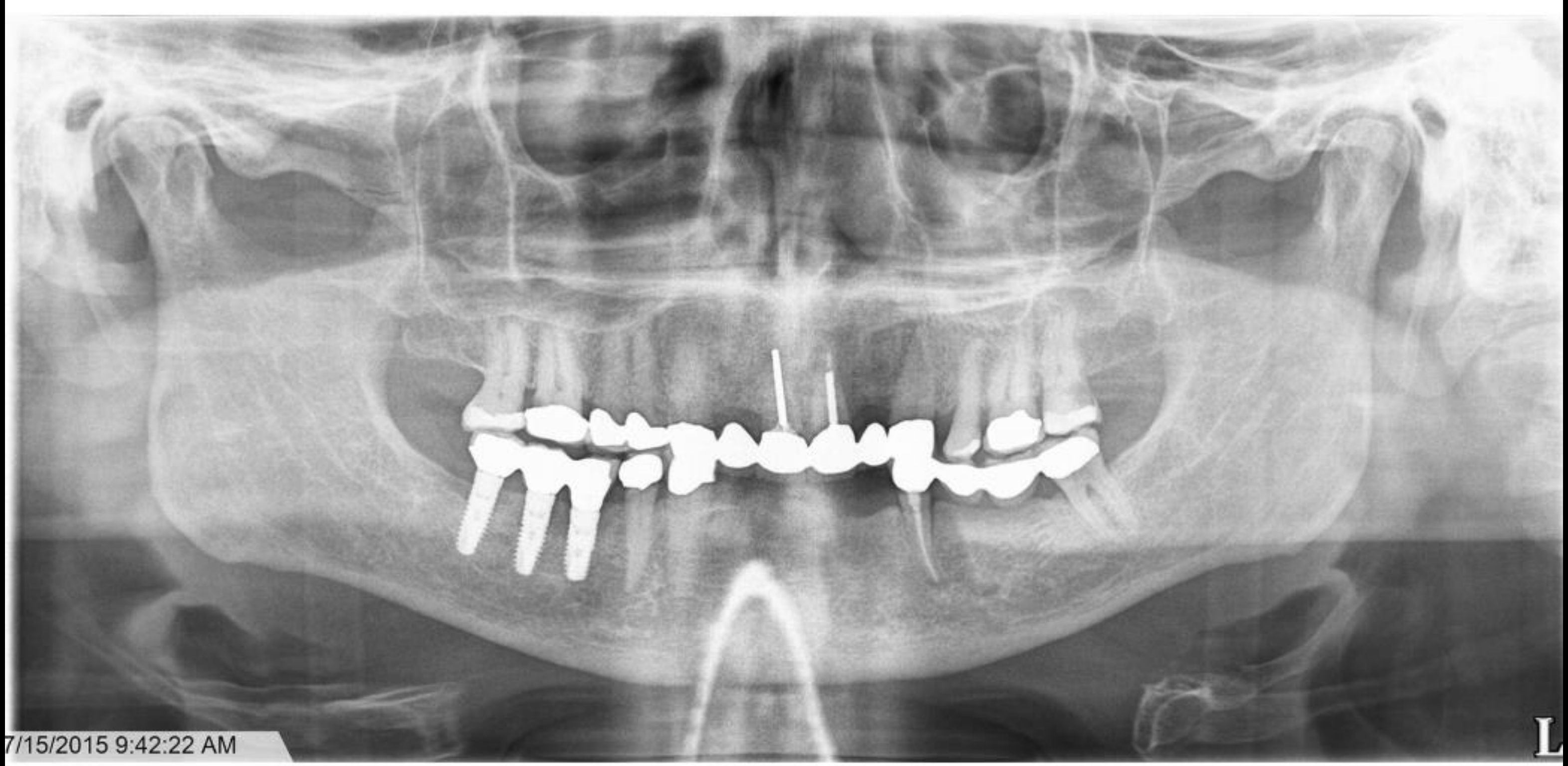


FIG 32-50 A metallic necklace appearing as a radiopaque loop in the region of the mandible.

Foreign Materials: Nose Ring

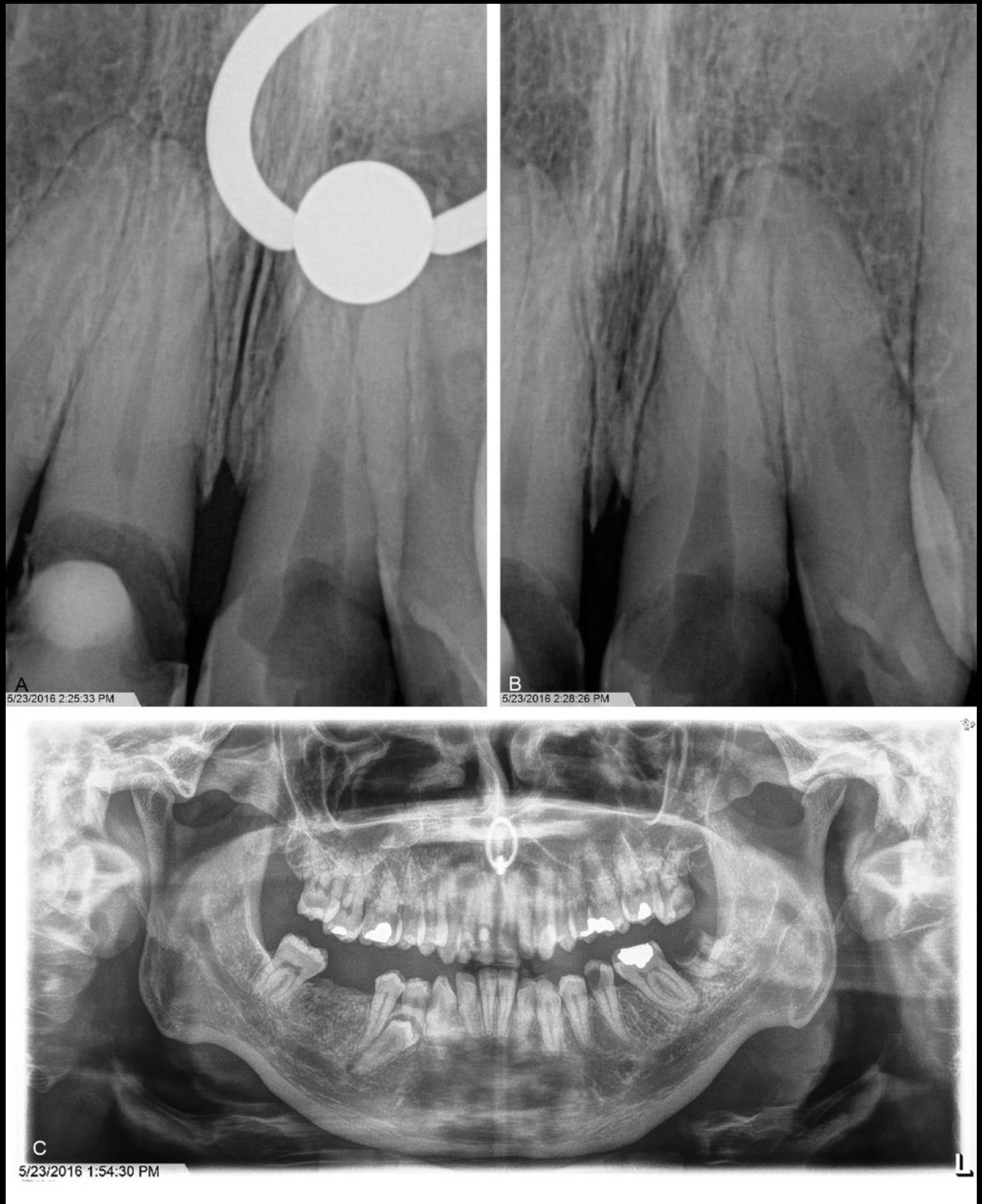


FIG 32-51 **A**, A maxillary anterior periapical image exposed with nose jewelry in place. **B**, The same exposure with the jewelry removed. **C**, A panoramic image of nose jewelry.

Foreign Materials: Eyeglasses

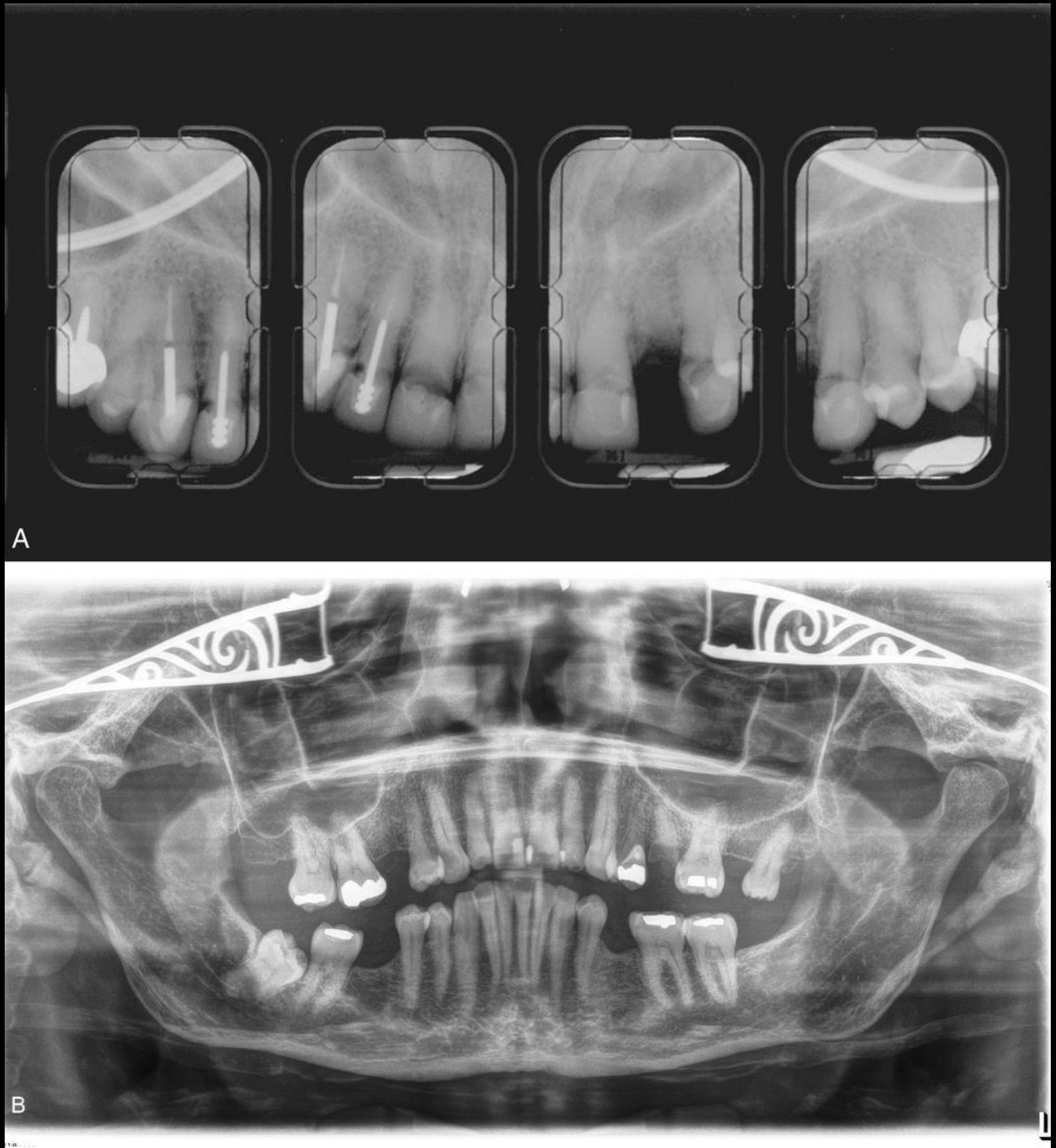


FIG 32-53 **A**, The rim of eyeglasses appear on both maxillary canine periapical images. **B**, The metal framework of the sides of eyeglasses seen on a panoramic image.